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PROJECT REPORT ENTITLED

DYNAMIC CONTEXT-AWARE TRIES FOR KNOWLEDGE GRAPH-CONSTRAINED LARGE LANGUAGE MODEL GENERATION

BY

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DECLARATION

We, BERNARD KIRK ADJANOR KATAMANSO, AMONOR ERICA, JOSEPH OSEI NYARKO and JESSICA AFUA ETORNAM NSAFOAH, declare that this project work is our own original work. It is being submitted for the degree of Bachelor of Science in Computer Science and Engineering at the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted, either in whole or in part, for any degree or examination at any other university.

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ABSTRACT

Large language models (LLMs) are capable of multi-hop reasoning but remain prone to hallucination. Knowledge graph-constrained decoding mitigates this by strictly restricting output tokens to valid paths within a knowledge graph (KG), ensuring absolute structural faithfulness unlike standard soft-grounding approaches (e.g., Retrieval-Augmented Generation). However, leading constrained paradigms—like Graph-Constrained Reasoning (GCR)—construct constraints solely from graph structure, ignoring the question's semantics and the evolving generation state.

This causes the *permissiveness problem*: at every decoding step, the allowed valid token set contains thousands of structurally accessible but contextually irrelevant paths. This flood of valid tokens forces the LLM to rely on the very parametric biases that constrained decoding was designed to suppress.

To resolve this, we propose Dynamic Context-Aware Tries (DCA-Trie), a novel framework that conditions the valid tokens jointly on the knowledge graph, the input question, and the partially generated reasoning chain. It upgrades the constraint oracle to an active semantic state $f(\mathcal{G}, E_q, q, y_{<t})$ by using a pretrained relevance scorer to filter candidate KG paths. Two architectures are developed: static semantic filtering applied at trie construction (DCA-Trie v1), and dynamic, step-wise bounding conditioned on the evolving generation (DCA-Trie v2).

Both variants are evaluated on the WebQSP and ComplexWebQuestions datasets over Freebase. Alongside standard metrics (Hits@1, F1), we introduce the Semantic Irrelevance Ratio (SIR)—a novel metric to uniquely isolate and quantify constraint permissiveness. To isolate and quantify constraint permissiveness, we introduce the Semantic Irrelevance Ratio (SIR), the first metric to measure the proportion of structurally valid but semantically irrelevant paths in the constraint set. Results demonstrate that DCA-Trie successfully eliminates semantically blind branches while preserving 100% structural faithfulness, charting a direct path toward more robust and efficient constrained reasoning.

DEDICATION

To our parents, who taught us that the truest things cannot be computed.

And to the machine, an elegant engine of probability, reminding us that predicting the word is not understanding the world.

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LIST OF ABBREVIATIONS

DCA-Trie	Dynamic Context-Aware Trie
LLM	Large Language Model
KGQA	Knowledge Graph Question Answering
RAG	Retrieval-Augmented Generation
KG	Knowledge Graph
GCR	Graph-Constrained Reasoning
CWQ	ComplexWebQuestions
RLHF	Reinforcement Learning from Human Feedback
CoT	Chain-of-Thought
DoG	Decoding on Graphs
SIR	Semantic Irrelevance Ratio
GCD	Grammar-Constrained Decoding
QA	Question Answering
NLP	Natural Language Processing
AI	Artificial Intelligence
BFS	Breadth-First Search

CHAPTER 1

INTRODUCTION

1.1 Background

Large language models (LLMs) have emerged as one of the most consequential advances in artificial intelligence, demonstrating capabilities across language understanding, generation, and multistep reasoning that were considered beyond reach only a decade ago (Brown *et al.*, 2020; OpenAI, 2024). Models such as the GPT series (Brown *et al.*, 2020; Achiam *et al.*, 2023) and the Llama family (Touvron *et al.*, 2023; Grattafiori *et al.*, 2024) achieve strong performance on a broad spectrum of language tasks by learning statistical regularities over vast corpora of text. The scale and breadth of this training enables these models to exhibit apparent knowledge across diverse domains, from history and science to legal reasoning and medical information, without any explicit representation of factual knowledge in their architecture.

Despite these achievements, a fundamental and persistent failure mode undermines the reliability of LLMs in knowledge-intensive settings: hallucination. Ji *et al.* (2023) define hallucination as the generation of content that is fluent and coherent but factually unsupported or contradicted by verifiable sources. Huang *et al.* (2023) provide a comprehensive taxonomy distinguishing intrinsic hallucination, where the output contradicts a provided source, from extrinsic hallucination, where the output cannot be verified against any source. Critically, hallucination is not a rare edge case. Luo *et al.* (2024) report that state-of-the-art retrieval-augmented approaches produce hallucinated reasoning steps in approximately one-third of cases even when granted access to a supporting knowledge graph. This failure rate renders unconstrained LLMs unsuitable for high-stakes knowledge-intensive tasks without additional mechanisms to enforce factual grounding. The structural inevitability of hallucination has been formally established in recent theoretical work. Xu *et al.* (2025) prove that for any computable ground truth function, no LLM training procedure can achieve zero hallucination probability across all inputs–hallucination is an inherent property of computable learning systems rather than a transient training artifact. This "inevitability" result underscores that mitigation strategies relying solely on improved training, scaling, or prompting cannot fully

eliminate hallucination; external structural constraints that ground generation in verifiable sources are theoretically necessary for guaranteed faithfulness. Knowledge graph-constrained decoding provides exactly such an external mechanism, making hallucination mathematically impossible rather than merely improbable.

The structural cause of hallucination is well understood. Autoregressive LLMs generate text by conditioning each token on all previously generated tokens, producing a probability distribution over the vocabulary at each step: $P(y_t \mid y_{<t}, x)$, where x is the input and $y_{<t}$ is the partially generated sequence (Brown *et al.*, 2020). This process is effective at producing locally coherent text but provides no mechanism for grounding each generated claim in a verifiable external source. An error introduced at step t propagates forward as the conditioning context for all subsequent tokens, producing confident and grammatically well-formed outputs whose factual incorrectness has no surface signal (Huang *et al.*, 2023; Ji *et al.*, 2023).

Knowledge graph question answering (KGQA) represents one of the most demanding contexts in which this failure mode manifests. A knowledge graph (KG) is a structured, human-curated database of verified facts encoded as (entity, relation, entity) triplets, for instance, (Marie Curie, award received, Nobel Prize in Physics) or (Paris, capital of, France), forming a large-scale graph over which multi-hop reasoning paths can be traced (Bollacker *et al.*, 2008). The Freebase knowledge graph, which provides the underlying resource for the standard KGQA benchmarks WebQSP (Yih *et al.*, 2016) and ComplexWebQuestions (Talmor and Berant, 2018), encodes tens of millions of such triplets covering a wide range of domains and relation types. Answering a multi-hop question such as 'What is the nationality of the director of the film in which the Blue Hawaii actor starred?' requires traversing a chain of three or more KG triplets in the correct order, a task for which an unconstrained LLM, relying entirely on parametric memory, may produce a plausible-sounding but factually incorrect chain that is indistinguishable in fluency from a correct one (Lan *et al.*, 2022; He *et al.*, 2021a).

The dominant paradigm for addressing this problem is retrieval-augmented generation (RAG), in which relevant passages or subgraphs are retrieved from an external source and provided to the LLM as additional context (Lewis *et al.*, 2020; Izacard and Grave, 2021). While RAG reduces hallucination by providing relevant source material, it does not prevent the LLM from ignoring that material or generating claims that go beyond it. The LLM remains free to

produce any token at any step; the retrieved context is a soft influence on its output, not a hard constraint on what it may generate (Asai *et al.*, 2024; Agrawal *et al.*, 2024b).

KG-constrained decoding addresses this limitation directly. Rather than providing the KG as a soft contextual influence, constrained decoding frameworks interpose a constraint oracle between the LLM's logit distribution and the token selection mechanism. At each generation step, the oracle computes a binary mask over the vocabulary, setting to zero the logit of any token that would not correspond to a valid continuation of a KG path. The masking formulation, introduced by Geng *et al.* (2023) in the context of grammar-constrained decoding, takes the form $\tilde{\alpha}_t = m_t \odot \alpha_t$, where α_t is the raw logit vector, $m_t \in \{0, 1\}^{|V|}$ is the validity mask, and $\odot$ denotes element-wise product. The generation process thus selects from a restricted set of valid tokens at every step, making structural hallucination, the generation of facts that do not exist in the KG, mathematically impossible rather than merely improbable (Willard and Louf, 2023; Geng *et al.*, 2023).

The leading implementation of this paradigm is Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2024), which achieves 92.6 Hits@1 on WebQSP and 75.8 Hits@1 on CWQ with verified 100% structural faithfulness. GCR constructs a KG-Trie, a prefix tree (Fredkin, 1960) over all KG paths within L hops of the question entities, obtained by breadth-first search, and uses it as the constraint oracle during beam-search decoding. Because every token in the generated reasoning chain must correspond to a valid prefix in the trie, the output is guaranteed to consist entirely of verified KG facts. This represents a qualitative improvement over both unconstrained LLMs and soft RAG approaches for the specific property of structural faithfulness.

However, a significant limitation remains. The KG-Trie in GCR is computed once, before generation begins, from the graph structure and the question entity set alone. It carries no information about the question semantics, the direction in which reasoning is proceeding, or which entities and relations have already been committed to in the partial generation $y_{<t}$. Formally, the valid token set W_{val} is a function $f(G, E_q)$, independent of q and $y_{<t}$, and remains identical across every generation step. For a question about a specific narrow fact, this trie may contain hundreds or thousands of structurally valid but semantically irrelevant paths, forcing the LLM to discriminate among them using its own parametric knowledge: precisely the hallucination-prone mechanism that constrained decoding was designed to

suppress (Li *et al.*, 2024; Luo *et al.*, 2024).

Decoding on Graphs (DoG) (Li *et al.*, 2024) introduces dynamic step-wise expansion, updating the constraint trie after each committed entity to include KG neighbors of the current node. This topological dynamism ensures the constraint set reflects the reasoning direction taken, addressing GCR's structural rigidity. DoG also employs question semantics, but crucially, only for *ranking* candidate paths during beam search–the valid token set itself remains determined solely by graph connectivity, admitting every neighbor regardless of relevance to the question or partial generation. The permissiveness problem therefore persists: thousands of structurally valid but semantically irrelevant paths remain in the valid set at each step, forcing the LLM to discriminate among them using parametric knowledge. Furthermore, DoG sacrifices the computational efficiency of pre-computed tries, rebuilding the constraint structure at each decoding step. These limitations motivate the DCA-Trie framework, which applies semantic filtering directly to the constraint oracle, ensuring that the valid token set at each step is both structurally faithful and semantically relevant.

1.2 Problem Statement

Existing KG-constrained decoding frameworks, including GCR (Luo *et al.*, 2024) and DoG (Li *et al.*, 2024), construct their constraint oracles as functions $f(G, E_q)$ of the knowledge graph and question entities, evaluated prior to or independently of the sequential generation process. This design has been shown to guarantee structural faithfulness (every generated token corresponds to a verified KG triplet), but provides no mechanism for adapting the constraint to the semantic content of the question or the direction taken by the partial generation (Geng *et al.*, 2023; Willard and Louf, 2023).

The consequence is a fundamental conflation of two distinct requirements. Structural validity requires that every generated token correspond to a valid KG path prefix, a requirement all existing frameworks satisfy. Contextual relevance requires that the constraint set reflect the semantic focus of the reasoning at the current step, a requirement no existing framework satisfies. As a result, existing constraint oracles are simultaneously too broad, admitting structurally valid but semantically irrelevant paths that force the model to rely on parametric discrimination, and too rigid, failing to narrow in response to the evolving reasoning direction as generation progresses (Li *et al.*, 2024; Luo *et al.*, 2024).

This project addresses the following specific problem: current KG-constrained decoding frameworks do not condition the valid token constraint on the input question semantics or the partial generation, leaving the LLM responsible for semantic discrimination among a large set of valid-but-irrelevant paths at every generation step. This residual reliance on parametric knowledge partially re-introduces the hallucination risk that constrained decoding was designed to eliminate, and is most pronounced in multi-hop settings where the number of valid paths grows exponentially with hop depth (Lan *et al.*, 2022; He *et al.*, 2021a; Talmor and Berant, 2018).

Very recent work has independently addressed the efficiency-faithfulness trade-off in KGQA. RouterKGQA (Yuan *et al.*, 2026) proposes routing between specialized and general models with semantic filtering at the answer level, achieving strong performance with minimal inference cost. While RouterKGQA and DCA-Trie both employ semantic relevance scoring, they operate at fundamentally different architectural stages: RouterKGQA filters *answers* after generation, whereas DCA-Trie filters *tokens* during generation. This distinction is critical: DCA-Trie's constraint oracle ensures the LLM never sees semantically irrelevant tokens, reducing cognitive load and enabling deterministic faithfulness guarantees that post-hoc filtering cannot provide. Furthermore, DCA-Trie introduces the Semantic Irrelevance Ratio (SIR; Chapter 3), the first metric to quantify constraint permissiveness independently of downstream accuracy–an evaluation dimension not addressed by routing-based approaches.

To address these limitations, this project proposes DCA-Trie, upgrading the constraint oracle from $W_{\text{val}}(t) = f(G, E_q)$ to $W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$. The upgraded framework must simultaneously satisfy three conditions: (i) structural faithfulness is preserved, every token admitted at each step corresponds to a valid KG path, as verified by exhaustive triplet checking against Freebase; (ii) semantic irrelevance is reduced, the proportion of the valid set consisting of paths irrelevant to the question decreases relative to GCR and DoG, as measured by the Semantic Irrelevance Ratio (SIR) introduced in this work; and (iii) answer recall is maintained: no gold answer path is erroneously excluded by the relevance filter, with false negative rate held below 5% on the WebQSP validation set.

1.3 Aim and Objectives

This project aims to design, implement, and evaluate a Dynamic Context-Aware Trie (DCA-Trie) framework for knowledge graph-constrained large language model generation, in which the valid token constraint at each decoding step is conditioned jointly on the knowledge graph structure, the input question semantics, and the partially generated reasoning chain, achieving improved answer accuracy on multi-hop KGQA benchmarks while maintaining 100% structural faithfulness to the underlying knowledge graph.

To achieve this aim, the following objectives have been set:

i. To characterize the permissiveness problem in existing KG-constrained decoding frameworks by defining and measuring the Semantic Irrelevance Ratio (SIR) of GCR's static KG-Tries on the WebQSP and CWQ benchmarks, stratified by hop depth.

ii. To design a semantic relevance scoring mechanism using pretrained sentence transformer embeddings (Reimers and Gurevych, 2019) that can filter candidate KG paths by joint relevance to the input question and partial generation at inference time, without task-specific fine-tuning.

iii. To implement a static semantic filtering variant (DCA-Trie v1) that applies the relevance scorer at KG-Trie construction time, reducing trie size before generation begins while maintaining a false negative rate below 5% on gold answer paths on the WebQSP validation set.

iv. To implement a step-wise dynamic expansion variant (DCA-Trie v2) in which the trie is updated after each committed triplet using the relevance scorer conditioned jointly on the question and the partial generation $y_{<t}$, directly operationalizing the $W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$ formulation.

v. To evaluate DCA-Trie v1 and DCA-Trie v2 against the GCR baseline and an unconstrained chain-of-thought (CoT) baseline on WebQSP and CWQ, measuring Hits@1, F1, structural faithfulness, SIR, and average trie size, with results stratified by question hop depth.

vi. To demonstrate the practical applicability of DCA-Trie by building an experimental interactive prototype system (web-based or command-line QA interface) for research

demonstration, allowing users to input complex natural language questions and receive factually grounded reasoning chains and answers, thereby showcasing the framework's real-world value beyond benchmark metrics.

1.4 Tools and Facilities Used

1.4.1 Software Tools

Table 1.1 presents an evaluation of the software tools implemented in this project, their role within the DCA-Trie framework, and the rationale for their selection.

Table 1.1 Implemented Software Tools Evaluation

Tool	Use in Project	Rationale
Python / HuggingFace Transformers	Model inference pipeline; loading and running GCR's Llama-3.1-8B checkpoint; integrating the relevance scorer	Provides reliable model loading, tokenization, and generation utilities, and enables direct use of the released GCR checkpoint.
GCR Framework	Primary baseline; KG-Trie construction, graph-constrained decoding, and result reproduction on WebQSP and CWQ	Provides a strong baseline and supports the trie-based constrained decoding mechanism that DCA-Trie extends.
Sentence-transformers / all-MiniLM-L6-v2	Semantic relevance scoring; computing cosine similarity between question-context and candidate KG path embeddings	Provides fast semantic embeddings without extra fine-tuning and supports efficient iterative decoding.
PyTorch	Deep learning framework underpinning all model inference and custom decoding logic	Supports implementation and testing of custom decoding logic and integrates smoothly with HuggingFace.
Freebase / WebQSP and CWQ Datasets	Knowledge graph source and KGQA benchmarks for evaluation	Provides standard datasets for this research area and enables direct comparison with prior work.
Google Colab Pro (A100, 40 GB)	GPU environment for all inference experiments; Llama-3.1-8B constrained decoding with beam search	Provides enough GPU memory for stable experiments and removes the need for local high-end hardware.
Visual Studio Code + Git / GitHub	Development environment and version control for all four team members	Supports day-to-day coding, team collaboration, version tracking, and reproducibility throughout the project.

1.4.2 Facilities

The following facilities were utilized in the execution of this project:

i. Google Colab Pro (A100, 40 GB VRAM) for all model inference and constrained decoding experiments.

ii. University library and online academic databases (ACM Digital Library, arXiv, Google Scholar, ACL Anthology) for literature review and citation verification.

iii. University internet infrastructure for cloud connectivity, dataset access, and retrieval of model checkpoints from the HuggingFace model hub.

iv. Shared team repository hosted on GitHub for version control, experiment logging, and reproducibility of all reported results.

1.5 Methods Used

The project employs the following methods:

i. Systematic review and critical analysis of the academic literature on constrained decoding, knowledge graph question answering, hallucination in LLMs, and retrieval-augmented generation.

ii. Formal gap analysis of the four foundational papers, GCR (Luo *et al.*, 2024), DoG (Li *et al.*, 2024), GCD (Geng *et al.*, 2023), and Outlines (Willard and Louf, 2023), to identify the specific limitations that motivate the DCA-Trie design, including the permissiveness problem and the absence of generation-state feedback in the constraint oracle.

iii. Formal specification of the DCA-Trie framework, including definition of the Semantic Irrelevance Ratio metric, the relevance scoring mechanism using sentence transformer embeddings (Reimers and Gurevych, 2019), the static filtering procedure for v1, and the step-wise dynamic expansion rule for v2.

iv. Reproduction of the GCR baseline on the WebQSP benchmark using GCR's publicly released Llama-3.1-8B checkpoint to establish a verified starting point within 1–2% of published Hits@1.

v. Implementation of DCA-Trie v1 and v2 as extensions to GCR's inference pipeline, with relevance threshold selected on a 100-question held-out validation subset of WebQSP to minimize false negative rate on gold answer paths.

vi. Quantitative evaluation of all system configurations on WebQSP and CWQ using Hits@1, F1, faithfulness rate, SIR, and average trie size per step, stratified by question hop depth following the evaluation protocol of Lan *et al.* (2022).

vii. Ablation studies examining the sensitivity of DCA-Trie's performance to the relevance threshold τ, the sentence encoder architecture, and beam size k, isolating the contribution of each mechanism to the overall result.

viii. Qualitative error analysis identifying failure cases in which DCA-Trie's semantic filter erroneously excludes a gold answer path, supporting honest characterization of the framework's limitations and providing concrete directions for future work.

1.6 Scope of Work

This project designs and evaluates a Dynamic Context-Aware Trie framework for KG-constrained LLM decoding, with the specific aim of addressing the permissiveness problem identified in GCR and DoG. The framework is developed and evaluated in the context of multi-hop KGQA over Freebase-grounded benchmarks (WebQSP and CWQ), which are the standard evaluation settings in the KG-constrained decoding literature. All experiments use GCR's publicly released Llama-3.1-8B checkpoint; no new model training or fine-tuning is performed. The semantic relevance scorer uses the all-MiniLM-L6-v2 sentence transformer without domain-specific adaptation. We emphasize that all experiments use GCR's publicly released Llama-3.1-8B checkpoint (Luo *et al.*, 2025), which includes their fine-tuned KG-specialized parameters. No additional model training or fine-tuning is performed in this work; our contribution is strictly the constraint oracle architecture and semantic filtering mechanism.

The scope is explicitly bounded as follows. Generalization of DCA-Trie to knowledge graphs other than Freebase, or to non-KGQA tasks such as entity linking or structured information extraction, is identified as future work and is not evaluated in this dissertation. Formal proofs of the faithfulness guarantee under dynamic pruning are beyond scope; empirical

measurement of the gold answer false negative rate is provided as evidence. The confidence-conditioned constraint tightening mechanism, in which constraint granularity is modulated by the LLM's token-level entropy, is described as a theoretical extension but is not implemented. Production-scale deployment, latency optimization for online serving, and integration with proprietary enterprise knowledge graphs are engineering extensions beyond the research scope of this project.

The primary contribution of this work is the DCA-Trie framework and the Semantic Irrelevance Ratio metric. Secondary contributions are the empirical characterization of constraint permissiveness in existing frameworks and the ablation analysis isolating the contribution of semantic filtering from dynamic expansion. All code, experimental configurations, and the curated evaluation datasets will be made publicly available to support reproducibility and future extension.

1.7 Organization of Work

Chapter 1 introduces the study by identifying the core problem, stating the project objectives, and defining the intended DCA-Trie solution, thereby establishing the motivation, scope, and overall direction of the research. Chapter 2 presents the literature review by examining prior work on hallucination in LLMs, KGQA, constrained decoding, and related frameworks, and it identifies key limitations in existing approaches to justify the need for a context-aware constraint mechanism. Chapter 3 combines problem formulation and methodology, defining the upgraded constraint setup, research questions, evaluation criteria, and implementation strategy for DCA-Trie v1 and v2 within the GCR pipeline (Luo *et al.*, 2024). Chapter 4 presents results and discussion, reporting quantitative evaluation across WebQSP and CWQ, analysing faithfulness and semantic irrelevance trends, and interpreting the findings through ablation and error analysis. Chapter 5 concludes the dissertation by summarizing the main contributions, highlighting practical implications, and presenting recommendations for future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Large Language Models (LLMs) have fundamentally transformed natural language processing by demonstrating unprecedented fluency and reasoning capabilities. However, their deployment in knowledge-intensive domains, where factual reliability is a strict prerequisite rather than a desirable trait, remains severely constrained. In contexts such as automated reasoning, medical diagnostics, and structured data querying, AI systems must do more than approximate plausible answers; they must ground their outputs in deterministically verifiable knowledge.

The foundational challenge hindering this adoption is hallucination: the generation of fluent but factually incorrect or unsupported claims. Because these models generate text autoregressively over an unconstrained vocabulary, they lack a built-in mechanism to ensure structural faithfulness to an external factual source. While current mitigation strategies, including chain-of-thought prompting and soft retrieval augmentation, significantly improve empirical accuracy, they fundamentally fail to provide hard, mathematical guarantees against hallucinated reasoning steps.

To address these challenges, this chapter examines existing research on these vulnerabilities and explores the progression of constrained decoding architectures designed to solve them. By doing so, it establishes the theoretical foundations for the Dynamic Context-Aware Trie (DCA-Trie). The review is structured around nine thematic pillars. First, Sections 2.2 and 2.3 formalize the problem by diagnosing hallucination's structural causes and exposing the faithfulness ceiling of unconstrained multi-step reasoning. Section 2.4 establishes the technical domain in which DCA-Trie operates: knowledge graph question answering and the mechanics of multi-hop inference. Sections 2.5 and 2.6 form the methodological core, tracing the evolution of constrained decoding paradigms, from foundational logit masking to leading graph-specific implementations like GCR and DoG, and analyzing their interaction with beam search. Section 2.7 examines retrieval-augmented generation to formally prove why soft grounding falls short of structural faithfulness, while Section 2.8 justifies the sentence

transformer embeddings used as the semantic relevance scorer at the heart of DCA-Trie. Finally, Sections 2.9 and 2.10 position this work against specific prior art and synthesize three unresolved research gaps that this project directly answers.

2.2 Hallucination in Large Language Models: Taxonomy, Structural Causes, and the Scaling Paradox

This section explores the root cause of the problem DCA-Trie sets out to solve. Rather than treating hallucination as a simple bug that can be patched, we examine it as a fundamental consequence of how autoregressive language models operate, meaning its elimination requires direct external constraints. To conclude, the section addresses a common counterargument, explaining why simply scaling up to larger models does not make these constraint mechanisms unnecessary.

2.2.1 Defining Hallucination: A Formal Taxonomy for Knowledge-Intensive Generation

Ji *et al.* (2023) define hallucination in natural language generation as the production of content that is fluent and internally coherent yet factually unsupported by, or in outright contradiction with, any verifiable source. Huang *et al.* (2023) refine this definition into a two-part taxonomy that is foundational to the present work. *Intrinsic hallucination* occurs when the model output directly contradicts material that is explicitly present in the provided source context. *Extrinsic hallucination* occurs when the output contains claims that cannot be verified against any provided or accessible external source, even if those claims are not outright contradictions.

When applied to the specific domain of knowledge graph question answering (KGQA), this two-part taxonomy translates into highly concrete, operational definitions. Intrinsic hallucination in KG-constrained generation means generating a triplet (e, r, e') that does not exist as a verified fact in the knowledge graph $\mathcal{G}$. Extrinsic hallucination means generating a multi-hop reasoning chain that cannot be traced to any verified path in $\mathcal{G}$, regardless of whether each individual generated token seems plausible in isolation. Both forms are unacceptable for safety-critical or knowledge-intensive applications, where factual reliability is a hard requirement rather than a desirable property. The severity of the problem is underscored by the observation in Luo *et al.* (2024) that even leading knowledge-graph-enhanced reasoning

methods, such as the RoG framework, still produce hallucinated reasoning steps in approximately 33% of cases. This single finding constitutes the strongest empirical justification for hard constraint mechanisms over soft grounding approaches.

2.2.2 The Autoregressive Generation Process as the Structural Root of Hallucination

Crucially, hallucination should not be viewed as a mere technical anomaly or a transient issue that can be easily trained away; rather, it is a fundamental architectural consequence of the autoregressive generation mechanism itself. At each generation step t, an autoregressive LLM computes a probability distribution over the full vocabulary conditioned on the input prompt x and all previously generated tokens $y_{<t} = (y_1, \ldots, y_{t-1})$:

$$P(y_t \mid y_{<t},\, x) \tag{2.1}$$

Three structural observations follow directly from this formulation (Brown *et al.*, 2020; Huang *et al.*, 2023; Ji *et al.*, 2023). First, no term in Equation (2.1) encodes a factual verification check: the distribution is computed entirely from learned weights and the conditioning context, with no mechanism for querying an external ground-truth source. Second, because $y_{<t}$ conditions all future steps, an error introduced at step $t-1$ propagates forward as part of the conditioning context for every subsequent token, producing grammatically fluent continuations whose factual incorrectness has no surface signal distinguishable from a correct chain. Third, the distribution is defined over the full vocabulary $\mathcal{V}$; every token has strictly positive probability at every step. The consequence is that structural hallucination is not a low-probability edge case; it is an ever-present possibility at every decoding step, and the fluency of the generated text provides no diagnostic signal that an error has occurred.

2.2.3 The Scaling Paradox: Why Larger Models Do Not Solve Hallucination

A natural and frequently encountered objection to constraint-based approaches is that sufficiently large models, or those trained with reinforcement learning from human feedback (RLHF), should be factually reliable enough to render hard constraints unnecessary. The evidence does not support this objection, and establishing this point is essential for a principled motivation of DCA-Trie. Lin *et al.* (2022) introduce TruthfulQA, a benchmark designed to measure whether models generate truthful answers by targeting questions that elicit common

human misconceptions. Their central finding is that larger models do not systematically outperform smaller models on truthfulness and that, for certain question categories, truthfulness accuracy actually decreases with scale, a phenomenon termed *inverse scaling*. This occurs because larger models natively become better at memorising and reproducing the false but highly prevalent misconceptions present in their training data.

As illustrated schematically in Figure 2.1, Truthfulness accuracy explicitly exhibits inverse scaling across base language models such as the GPT-3 and GPT-Neo families. Unconstrained language modeling performance fails to improve monotonically with parameter count and instead regresses robustly into the inverse-scaling region. By contrast, models explicitly fine-tuned on QA tasks (such as UnifiedQA) exhibit positive scaling, demonstrating that scaling standard autoregressive imitation alone inherently penalizes factual reasoning.

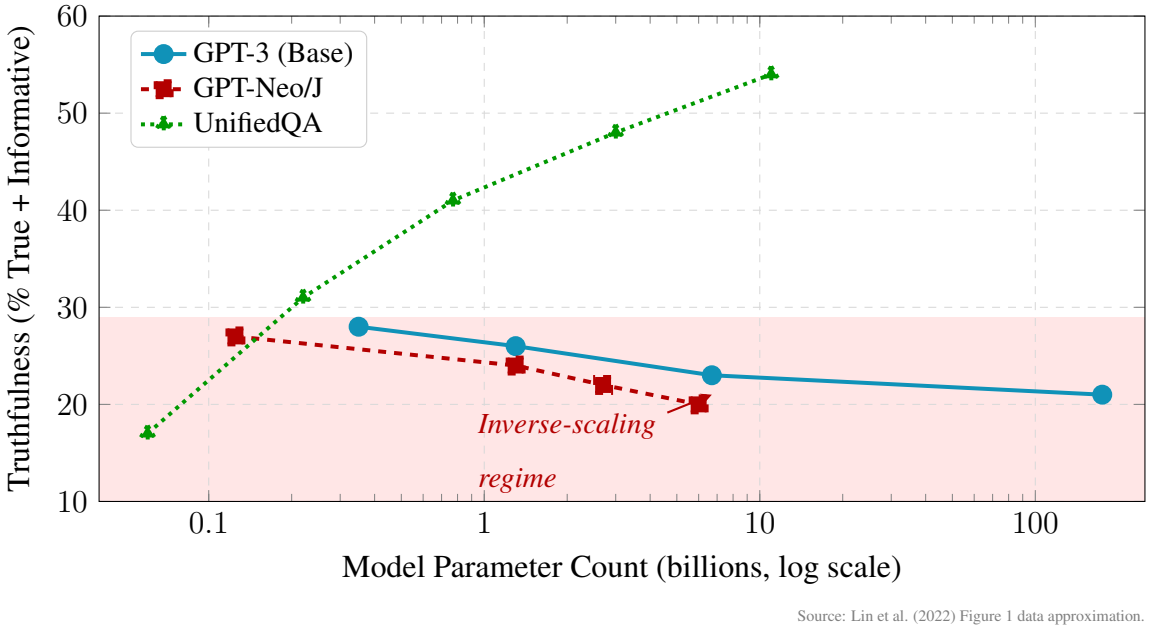

Figure 2.1 Relationship between model scale and truthfulness on the TruthfulQA benchmark.

The sycophancy problem identified by Perez *et al.* (2023) compounds this picture. RLHF-tuned models are trained to produce outputs aligned with user preferences; since users tend to rate confident and fluent responses favorably, larger RLHF-tuned models are more likely to generate confident-sounding but factually incorrect statements that match expected answers rather than true ones. Confidence and accuracy are thus decoupled by the very training procedure that was supposed to improve model reliability. Kandpal *et al.* (2023) identify a further limitation through the memorisation-generalisation tension: larger models memorize

more training data and therefore perform better on factual recall tasks involving frequently seen entity combinations, but they do not improve on compositional multi-hop questions that require systematic reasoning over entity combinations unseen during training. that require systematic reasoning over entity combinations unseen during training. Knowledge graph question answering is precisely this kind of task: the correct answer to a novel three-hop question requires chaining entities and relations in a configuration that almost certainly did not appear as a single training example. Mallen *et al.* (2023) corroborate this by showing that even GPT-4 hallucinates at a non-trivial rate on questions involving tail entities with low training-set frequency, regardless of the model's overall capability.

These findings collectively demonstrate that hallucination in large language models is not a limitation of scale, but a fundamental structural flaw. Autoregressive generation inherently lacks a mechanism to ground generated tokens in verifiable external sources, an absence that cannot be resolved simply by increasing parameter counts. The following sections review approaches that attempt to address this limitation, progressing from soft grounding via retrieval augmentation to the hard constraint mechanisms that motivate this dissertation.

2.3 Chain-of-Thought Reasoning, Self-Consistency, and the Faithfulness Ceiling of Unconstrained Multi-Step Generation

A natural question often arises when evaluating large language models: if chain-of-thought prompting already yields such significant improvements in multi-step reasoning, why do we need constrained decoding at all? The answer developed in this section is that while chain-of-thought and its extensions improve reasoning *accuracy in expectation*, they cannot offer a *structural faithfulness guarantee*, the exact property that DCA-Trie is designed to provide. Recognizing this distinction is essential, as it highlights why prompting alone is insufficient and why constrained decoding remains fundamentally necessary.

2.3.1 Chain-of-Thought Prompting: Eliciting Step-by-Step Reasoning from LLMs

Chain-of-thought (CoT) prompting, introduced by Wei *et al.* (2022), is the dominant paradigm for improving multi-step reasoning in large language models. The mechanism is straightforward: by including worked examples in the prompt that show intermediate reasoning steps leading to the final answer, the model learns to produce its own intermediate steps before

giving a final answer to a new question. This shifts the model from direct answer prediction to the decomposition of complex problems into a chain of simpler sub-problems, each resolved in sequence. Wei *et al.* (2022) show that this ability is emergent: it is negligible in models below approximately 100 billion parameters and arises sharply beyond this threshold. Empirically, CoT prompting yields substantial improvements on mathematical reasoning benchmarks such as GSM8K and MATH, as well as on multi-hop question answering tasks, in comparison with direct prompting baselines.

Formally, a CoT output for a given question x is structured as a sequence of intermediate reasoning steps followed by a final answer:

$$\big(r_1,\, r_2,\, \ldots,\, r_k,\, a\big) \tag{2.2}$$

where r_i denotes each intermediate reasoning step and a is the final answer. The language model generates each step autoregressively, using the same probabilistic mechanism described in Equation (2.1):

$$P(r_i \mid r_{<i},\, x)$$

The intrinsic limitation of this approach is immediately apparent: there is no mechanism to ensure that any given step r_i corresponds to an actual, verified triplet within the knowledge graph $\mathcal{G}$. If the model hallucinates an intermediate step r_i, that error becomes part of the conditioning context for the next step r_{i+1}, cascading forward and irreparably compromising the final answer a. As a result, a CoT sequence can read as entirely fluent, coherent, and plausible while still containing one or more steps that lack any basis in verified facts.

2.3.2 Self-Consistency and Tree-of-Thought: Improving Reliability Without Improving Faithfulness

Two major extensions of chain-of-thought attempt to address these reliability limitations, yet neither fundamentally resolves the structural faithfulness problem. The first, self-consistency (Wang *et al.*, 2023), combats the high variability of individual CoT chains by sampling multiple independent reasoning paths and selecting the most frequent final answer (majority voting). This approach effectively marginalizes out anomalous reasoning errors and substantially boosts accuracy on benchmarks like GSM8K and multi-hop QA. However, it still falls

short of providing a hard guarantee: there is no assurance that any single sampled chain is fully grounded in verified knowledge graph facts. It is entirely possible — especially for deep, multi-hop questions where error probabilities compound — for a majority of independently generated chains to all suffer from hallucinations, resulting in an incorrect majority-vote answer.

The second major extension, Tree-of-Thought (Yao *et al.*, 2023), takes a more structured approach by framing the generation process as a search over a tree of partial reasoning steps. It uses the language model as both the generator of new steps and the evaluator that scores and prunes unpromising branches. This deliberate, planning-oriented approach significantly improves performance on complex tasks that require backtracking. Ultimately, however, Tree-of-Thought operates entirely within the model's own parametric knowledge: the node scoring and pruning decisions are made by the same system generating the steps, without ever querying an external, verified knowledge graph.

The decisive observation for both of these prompting-based methods is that they can only reduce the *statistical probability* of hallucinated reasoning; they cannot actually guarantee a zero-hallucination output. This is because neither method alters the fundamental token-level generation mechanism. At any given step, the valid token set remains the entire vocabulary $\mathcal{V}$, and every token retains a strictly positive probability of being selected. This highlights the critical distinction introduced by constrained decoding, which enforces absolute structural faithfulness by setting the probability of every invalid token to exactly zero.

Table 2.1 Comparison of chain-of-thought family methods and KG-constrained decoding.

Method	**Multi-hop Acc.**	**Faithfulness Guarantee**	**Ext. Verification**	**Parametric Reliance**
Chain-of-Thought (Wei *et al.*, 2022)	High	No	No	High
Self-Consistency (Wang *et al.*, 2023)	High	No	No	High
Tree-of-Thought (Yao *et al.*, 2023)	High	No	No	High
KG-Constrained Decoding	High	Yes	Yes	Low

2.3.3 The Faithfulness Ceiling: The Mathematical Limits of Prompting

The fundamental inadequacy of prompting arises directly from its probabilistic formulation. All prompting strategies, whether relying on few-shot examples, chain-of-thought decomposition, self-consistency aggregation, or tree-structured search, operate strictly by modifying the

input provided to the language model. While this favorably shifts the probability distribution toward correct sequences, it leaves the underlying autoregressive generation mechanism entirely untouched. Because decoding still occurs over the unrestricted vocabulary $\mathcal{V}$, every token technically retains a strictly positive probability of being sampled at any given step.

True structural faithfulness, defined as the absolute guarantee that every generated token forms part of a verified knowledge graph path, requires that all invalid tokens be assigned a probability of *exactly* zero. This cannot be accomplished at the prompt level; it necessitates a fundamental intervention in the decoding algorithm itself, which is precisely the function of constrained decoding (Geng *et al.*, 2023; Willard and Louf, 2023).

As formally distinguished by Geng *et al.* (2023) and Willard and Louf (2023), prompting acts as a *soft influence*, whereas constrained decoding operates as a *hard mechanism*. While intelligent prompting can significantly elevate factual accuracy in expectation, only constrained decoding can guarantee structural faithfulness with a probability of one.

2.4 Knowledge Graph Question Answering: Structured Reasoning, Benchmark Design, and the Exponential Complexity of Multi-Hop Inference

This section establishes the technical domain in which the DCA-Trie framework operates. The principles and structural constraints introduced here provide the foundational rationale for the architectural design decisions detailed in Chapter 3.

2.4.1 Knowledge Graphs as Structured Repositories of Verified Facts: Formal Definition and Properties

A knowledge graph is formally defined as a directed labelled graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$, where $\mathcal{E}$ is the set of entities, $\mathcal{R}$ is the set of relation types, and $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ is the set of verified factual triplets. A standard concrete example from Freebase (Bollacker *et al.*, 2008) is the triplet (*Marie Curie*, *award_received*, *Nobel_Prize_in_Physics*), which logically connects two entities via a directed relation.

Freebase encodes tens of millions of such triplets spanning geography, history, entertainment, and science. It serves as the underlying factual resource for both benchmark datasets evaluated in this work: WebQSP (Yih *et al.*, 2016) and ComplexWebQuestions (Talmor and

Berant, 2018). Given an initial entity e_0, a multi-hop reasoning path of length L within $\mathcal{G}$ is represented as the sequence:

$$p = (e_0,\, r_1,\, e_1,\, r_2,\, e_2,\, \ldots,\, r_L,\, e_L) \tag{2.3}$$

where each consecutive step $(e_{i-1},\, r_i,\, e_i)$ forms a valid triplet within $\mathcal{T}$. In the context of KGQA, the correct answer to a multi-hop question q corresponds to the terminal entity e_L of a specific, structurally sound path p. This formalization is load-bearing throughout this dissertation: the structural faithfulness guarantee encoded by DCA-Trie strictly mandates that every generated sequence inherently forms a valid prefix of some defined path p within $\mathcal{G}$.

The critical distinction between a structured KG and an unstructured text corpus is that every triplet in $\mathcal{T}$ is human-verified, machine-readable, and unambiguously delineated. This innate topology enables KG-constrained generation to operate in a manner entirely distinct from standard retrieval augmentation (RAG). Rather than relying on a probabilistic semantic similarity search against raw text, a constraint oracle can systematically verify any candidate token against $\mathcal{G}$ via precise, deterministic graph lookups.

Concrete Example: A Knowledge Graph Subgraph from Freebase

To make the formal definition concrete, consider a small subgraph extracted from Freebase. Figure 2.2 illustrates a directed graph with six entity nodes connected by five labelled relation edges. Each edge represents a single verified triplet $(e, r, e') \in \mathcal{T}$. For instance, the triplet (*Blue_Hawaii*, *film.film.featured_film_location*, *Hawaii*) encodes the fact that the film "Blue Hawaii" was filmed in Hawaii. A valid multi-hop reasoning path through this subgraph might be:

(*Blue_Hawaii*, *film.film.featured_film_location*, *Hawaii*, *location.location.adjoining_water*, *Pacific_Ocea*

This path has length $L = 2$ (two hops) and terminates at the entity *Pacific_Ocean*. If a question asks "What large body of water adjoins the location where Blue Hawaii was filmed?", the correct answer corresponds to the terminal entity of this verified path. The structural faithfulness guarantee of DCA-Trie ensures that every intermediate step of the

model's generated reasoning chain corresponds to an actual verified triplet in $\mathcal{T}$ making hallucinated reasoning steps mathematically impossible.

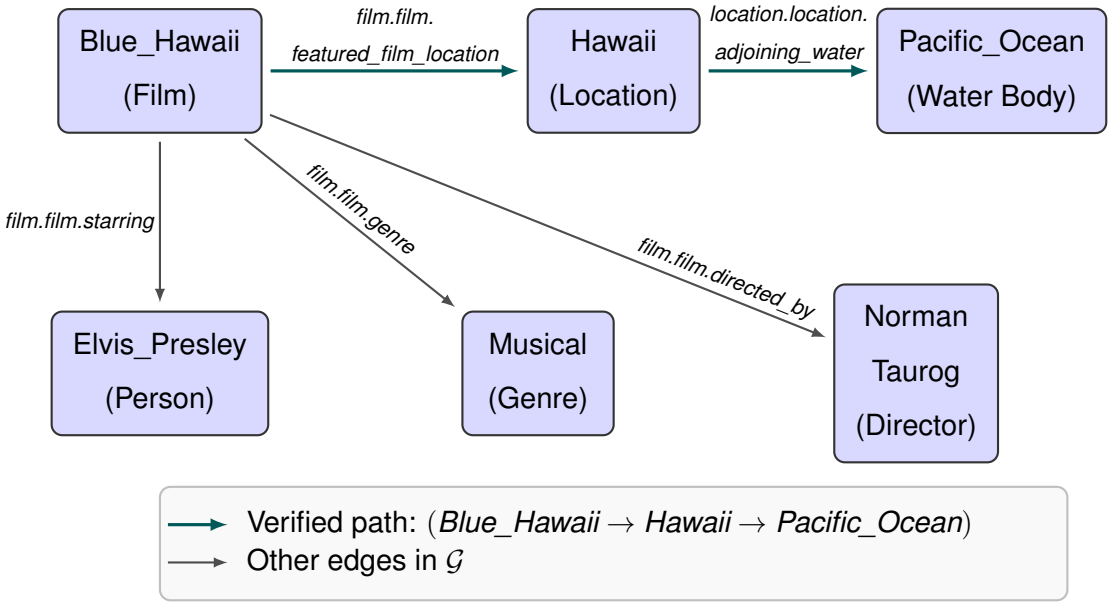

Figure 2.2 Example knowledge graph subgraph from Freebase

2.4.2 WebQSP and ComplexWebQuestions: Benchmark Architecture, Evaluation Protocol, and Hop-Depth Stratification

Two benchmark datasets provide the empirical evaluation framework for this dissertation. WebQSP (Yih *et al.*, 2016) is derived from the WebQuestions dataset and comprises 4,737 questions grounded in Freebase, with reasoning depths of one to two hops. Each question is annotated with its corresponding SPARQL query against Freebase and is typically evaluated using Hits@1 and F1 scores. Within the context of constrained generation, WebQSP represents a low-complexity verification regime; because the valid path set branching from any question entity is strictly bounded, the target path is relatively straightforward to discriminate from incorrect alternatives.

ComplexWebQuestions (CWQ; Talmor and Berant 2018) extends this paradigm by applying systematic transformations – specifically composition, conjunction, and comparative or superlative operations – to WebQSP seed queries. This yields approximately 34,000 questions requiring deeper reasoning chains spanning up to four hops. For generative frameworks, CWQ represents the combinatorial frontier where the *permissiveness problem* is most acute. At a depth of four hops, the set of structurally valid paths reachable from the question entities grows exponentially. Forced to discriminate among thousands of formally valid but

semantically irrelevant paths relying solely on its internal parametric knowledge, the LLM is exposed to the exact hallucination failure mode that DCA-Trie targets. To systematically measure this degradation, the hop-depth-stratified evaluation protocol from Lan *et al.* (2022) is adopted for the experiments in Chapter 4.

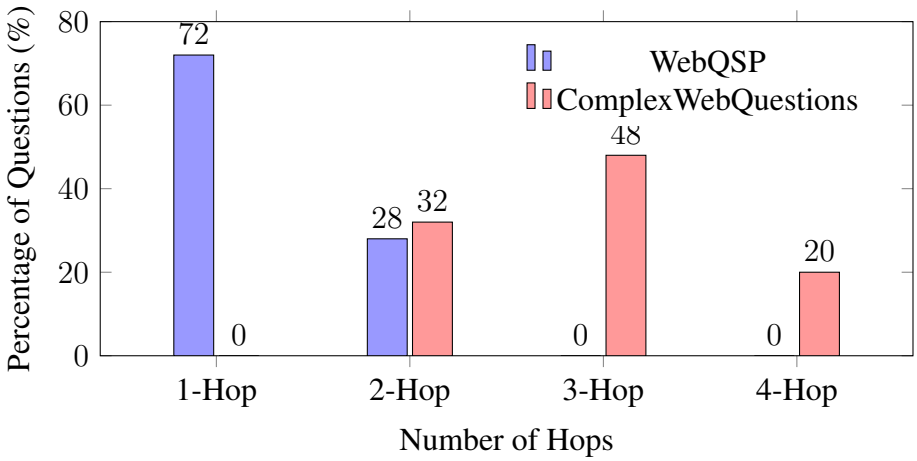

Figure 2.3 Hop-depth distributions for WebQSP and ComplexWebQuestions.

2.4.3 Multi-Hop Reasoning Complexity: Exponential Path Growth and the Permissiveness Problem Quantified

The qualitative difference between one-hop and multi-hop questions can be made precise through a combinatorial analysis. A one-hop question requires selecting a single correct relation from the set of outgoing edges of the question entity e_q. The valid candidate set is bounded by the out-degree d of e_q. A multi-hop question of depth L requires selecting a correct sequence of L consecutive steps, where at each step the valid set multiplies. The upper bound on the number of valid KG paths of length L reachable from the question entities $\mathcal{E}_q$ is:

$$|\mathcal{P}_L| \;\leq\; |\mathcal{E}_q| \times d^L \tag{2.4}$$

where d represents the average out-degree of entities in $\mathcal{G}$. Assuming a conservative average out-degree $d \approx 20$ for well-connected entities in Freebase, and a typical query linking to $|\mathcal{E}_q| \in \{1, 2, 3\}$ seed entities, we can instantiate Equation (2.4): at $L = 2$, the upper bound is $3 \times 400 = 1{,}200$ valid paths; at $L = 3$, the bound grows to $3 \times 8{,}000 = 24{,}000$ valid paths;

at $L = 4$, the bound reaches $3 \times 160{,}000 = 480{,}000$ valid paths. In each case, exactly one path is the correct targeted answer.

This exponential growth formally quantifies the permissiveness problem inherent to GCR and DoG. When the constraint oracle relies purely on structural validity, $f(\mathcal{G}, \mathcal{E}_q)$, the valid token set at every generation step is flooded with thousands of structurally permitted but semantically irrelevant paths. The LLM is consequently forced to discriminate among them relying solely on its internal parametric knowledge – precisely the hallucination-prone failure mode that constrained decoding was originally designed to suppress (Luo *et al.*, 2024; Li *et al.*, 2024). The Semantic Irrelevance Ratio (SIR), introduced in Chapter 3 of this dissertation, provides the first direct measure of this phenomenon. Empirical evidence for performance degradation as a function of hop depth is documented in He *et al.* (2021b) and Lan *et al.* (2022).

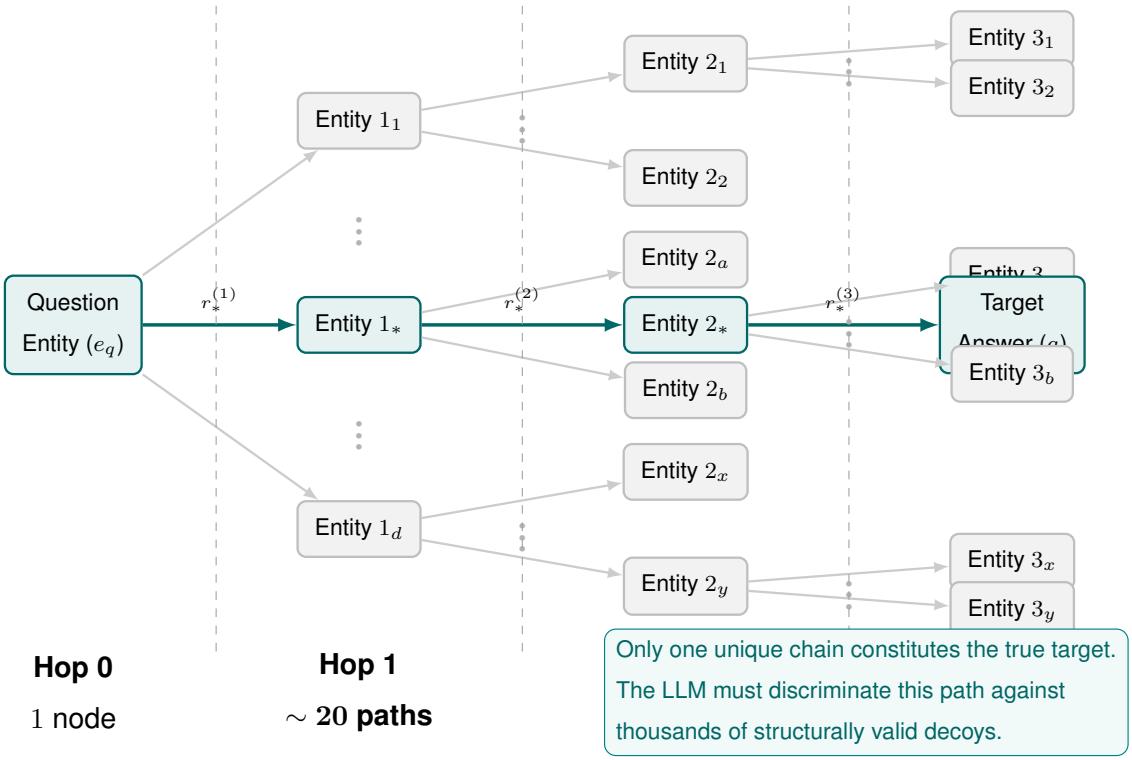

Figure 2.4 Exponential growth of structurally valid KG paths with hop depth.

2.5 Constrained Decoding for Faithful Text Generation: From Logit Masking to Knowledge Graph Oracles

This section establishes the technical lineage of constrained decoding paradigms leading to the proposed framework. It progresses from the foundational logit-masking mechanism through grammar-based constraints to contemporary knowledge-graph-specific oracles. Each framework reviewed serves either as a direct methodological baseline or as a foundational model whose specific structural limitations the DCA-Trie architecture is designed to resolve.

2.5.1 The Logit-Masking Paradigm: Hard Constraints on Autoregressive Vocabulary Selection

The logit-masking paradigm is the foundational mechanism for hard-constrained generation (Geng *et al.*, 2023; Willard and Louf, 2023). The key insight is that rather than influencing the probability distribution over tokens, as prompting does, a constraint oracle computes a binary mask that sets the logit of every invalid token to negative infinity (effectively zero probability after softmax) before the token selection step. This renders the generation of invalid tokens mathematically impossible rather than merely improbable.

Formally, let $\boldsymbol{\alpha}_t \in \mathbb{R}^{|\mathcal{V}|}$ be the raw logit vector at decoding step t, and let $\mathbf{m}_t \in \{0,1\}^{|\mathcal{V}|}$ be the binary validity mask computed by the constraint oracle. The masked logit vector is:

$$\tilde{\boldsymbol{\alpha}}_t \;=\; \mathbf{m}_t \odot \boldsymbol{\alpha}_t \tag{2.5}$$

where $\odot$ denotes element-wise product (with the convention that $0 \odot \alpha = -\infty$ before softmax). Token selection under greedy decoding is $y_t = \arg\max(\tilde{\boldsymbol{\alpha}}_t)$. Under beam search with beam width k, the top-k partial sequences are maintained and ranked by cumulative log-probability of masked logits.

The central research question of this dissertation follows directly from this formulation: *how should* $\mathbf{m}_t$ *be computed?* The answer determines the expressiveness, faithfulness, and efficiency of the entire framework. GCR computes $\mathbf{m}_t$ from a function $f(\mathcal{G}, \mathcal{E}_q)$ evaluated once before generation. DCA-Trie computes $\mathbf{m}_t$ from a function $f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$ updated at every step. Every subsequent subsection is a step in the progression toward this answer.

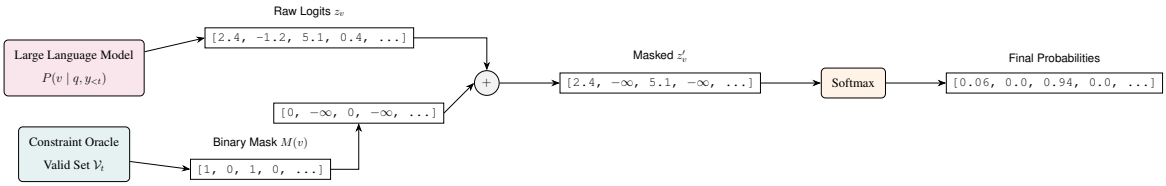

Figure 2.5 The logit-masking mechanism. Token selection operates exclusively over the valid, non-zero subset defined by the constraint oracle mask.

2.5.2 Grammar-Constrained Decoding: GCD and Outlines as Foundational Format Constraints

Geng *et al.* (2023) introduce Grammar-Constrained Decoding (GCD), which constrains generation to strings conforming to a context-free grammar by using an Earley parser as the constraint oracle. Willard and Louf (2023) introduce Outlines, which achieves the same goal for regular-expression and JSON-schema constraints via finite-state automata. Both works establish the critical efficiency insight that the constraint structure can be precomputed offline, as parse states or automaton transitions, and queried in amortised constant time at each inference step. This precomputation insight is central to DCA-Trie v1, which preserves it through static semantic filtering at trie construction time.

Algorithm 1 Grammar-Constrained Decoding Algorithm

Require: Context-free grammar $\mathcal{G}_{\text{CFG}}$; LLM with vocabulary $\mathcal{V}$; prompt x

Ensure: Output sequence $y_1 \dots y_T$ valid under $\mathcal{G}_{\text{CFG}}$

1: **Precompute:** Earley parse automaton $\mathcal{A}$ from $\mathcal{G}_{\text{CFG}}$
2: **for** $t = 1$ **to** T **do**
3: $\quad \mathcal{V}_t \leftarrow \{v \in \mathcal{V} : y_{<t} \| v \text{ is a valid grammar prefix under } \mathcal{A}\}$ ▷ Oracle query
4: $\quad \mathbf{m}_t[v] \leftarrow \mathbf{1}[v \in \mathcal{V}_t]$ for all $v \in \mathcal{V}$
5: $\quad \tilde{\boldsymbol{\alpha}}_t \leftarrow \mathbf{m}_t \odot \textsc{LLMLogits}(y_{<t}, x)$
6: $\quad y_t \leftarrow \arg\max(\tilde{\boldsymbol{\alpha}}_t)$
7: **end for**
8: **return** $y_1 \dots y_T$

Figure 2.6 Grammar-Constrained Decoding Algorithm.

The shared limitation of GCD and Outlines is stated precisely: both frameworks constrain the *format* of the output (grammar, schema, structure) but not the *factual content*. A grammatically valid JSON response can contain entirely hallucinated facts. As annotated in Figure 2.6, the valid set $\mathcal{V}_t$ depends only on the partial string and the grammar; there is no dependence on question semantics or on a verified factual source. This is the same structural limitation present in GCR's KG-Trie, which constrains to KG path format while remaining semantically blind to the question. The transition from format constraints to knowledge-graph content constraints is the next step in the progression.

2.5.3 Graph-Constrained Reasoning (GCR): Structural Faithfulness via KG-Trie Oracles

Graph-Constrained Reasoning (GCR; Luo *et al.* 2024) is the primary baseline for this dissertation and represents the state of the art in KG-constrained decoding at the time of this work. GCR introduces five components. First, *entity linking* identifies the question entity set $\mathcal{E}_q$ from the natural language question before generation begins. Second, *KG-Trie construction* performs a breadth-first search from $\mathcal{E}_q$ up to L hops, encoding all reachable KG paths as a prefix tree. A trie node at depth d corresponds to a partial path of d tokens. Third, *constrained decoding* queries the trie at each generation step with the current partial generation $y_{<t}$ to return the set of tokens that extend a valid trie prefix. Fourth, the *faithfulness guarantee* follows: because every token must correspond to a valid trie prefix, and every trie path corresponds to a verified triplet chain in $\mathcal{T}$, structural hallucination is mathematically impossible. Fifth, GCR achieves 92.6 Hits@1 on WebQSP and 75.8 Hits@1 on CWQ with verified 100% structural faithfulness using a Llama-3.1-8B backbone (Luo *et al.*, 2024).

Formally, the valid token set under GCR is:

$$\mathcal{W}_{\text{val}}^{\text{GCR}} \;=\; f(\mathcal{G},\, \mathcal{E}_q) \tag{2.6}$$

The KG-Trie $\mathcal{T}_{\text{trie}}$ is constructed as $\mathcal{T}_{\text{trie}} = \text{BFS}(\mathcal{G}, \mathcal{E}_q, L)$. At each step t:

$$\mathcal{V}_t = \left\{ v \in \mathcal{V} : \text{prefix}(y_{<t}) \| v \in \mathcal{T}_{\text{trie}} \right\}$$

The critical annotation is that $\mathcal{W}_{\text{val}}^{\text{GCR}}$ has *no dependence* on q (the question text) or on $y_{<t}$ (the partial generation state). The valid set is semantically blind and computationally

fixed before generation begins. This is the exact formulation that DCA-Trie upgrades to $\mathcal{W}_{\text{val}}^{\text{DCA}}(t) = f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$.

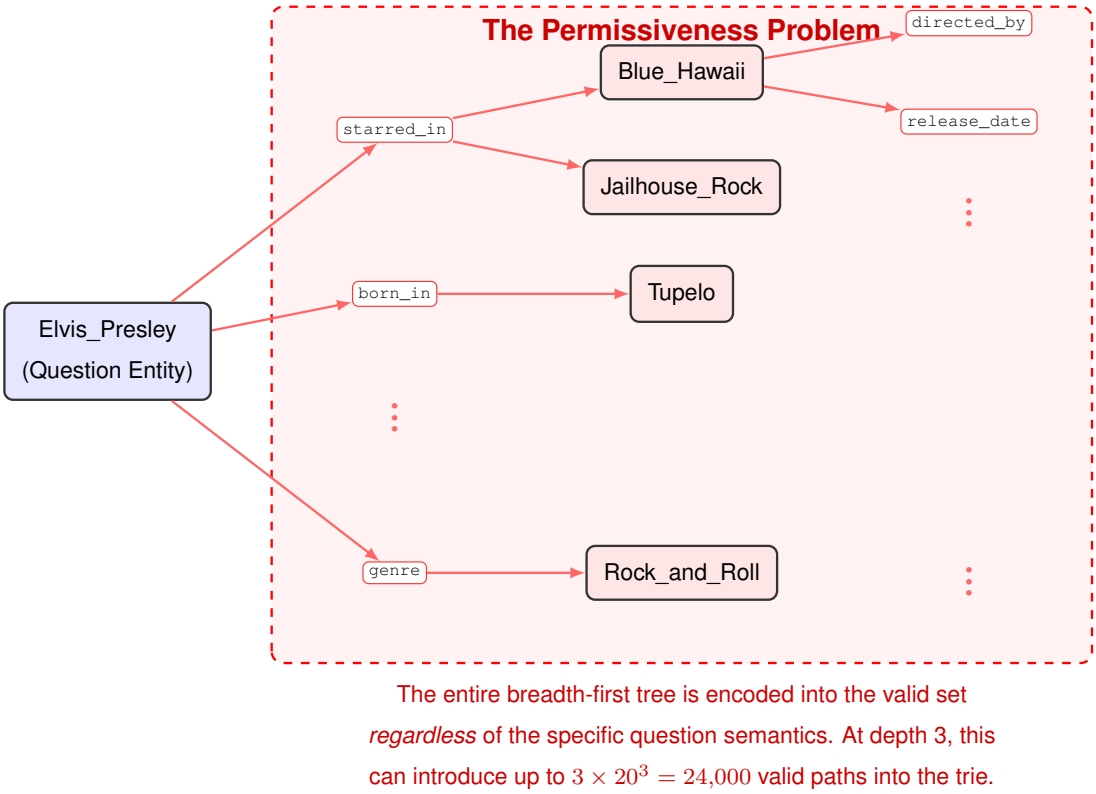

Figure 2.7 The GCR KG-Trie illustrating the permissiveness problem.

2.5.4 Decoding on Graphs (DoG): Step-Wise Topological Expansion and Its Persistent Limitations

Li *et al.* (2024) introduce Decoding on Graphs (DoG) as a partial remedy to GCR's static trie. DoG introduces step-wise trie expansion: after committing to an entity e_t during generation, the valid set is extended with all KG neighbors of e_t. Formally, after committing entity e_t:

$$\mathcal{T}_{t+1} \;=\; \mathcal{T}_t \;\cup\; \big\{(e_t,\, r,\, e') : (e_t,\, r,\, e') \in \mathcal{T} \text{ for all } r, e'\big\} \tag{2.7}$$

This makes the constraint topologically dynamic: it grows as reasoning progresses through the graph. DoG offers two genuine improvements over GCR. First, the valid set at step t reflects the entities already committed to in the reasoning chain, so the trie cannot contain paths that are structurally inconsistent with the committed reasoning direction. Second, the

dynamic expansion can handle knowledge graphs too large to enumerate statically within the pre-generation BFS.

However, DoG has two persistent limitations. First, as is explicit in Equation (2.7), the expansion rule is semantically blind: every neighbor of every visited entity is admitted to $\mathcal{T}_{t+1}$ regardless of its relevance to q or to the partial generation $y_{<t}$. The permissiveness problem therefore persists: the number of paths in the valid set may remain in the thousands at every step. Second, DoG sacrifices the computational efficiency of precomputed trie structures identified by Willard and Louf (2023): because the trie must be rebuilt at each step, the amortised per-step query cost of precomputed structures is forfeited.

DCA-Trie addresses both limitations simultaneously. The DCA-Trie v2 expansion rule replaces the semantically blind expansion of Equation (2.7) with a semantically conditioned rule (defined formally in Chapter 3):

$$\mathcal{T}_{t+1} = \mathcal{T}_t \cup \left\{ (e_t,\, r,\, e') : \textsc{score}(e',\, q,\, y_{<t}) \geq \tau \right\}$$

The single structural change, conditioning expansion on a semantic relevance score, is the core algorithmic contribution of DCA-Trie v2.

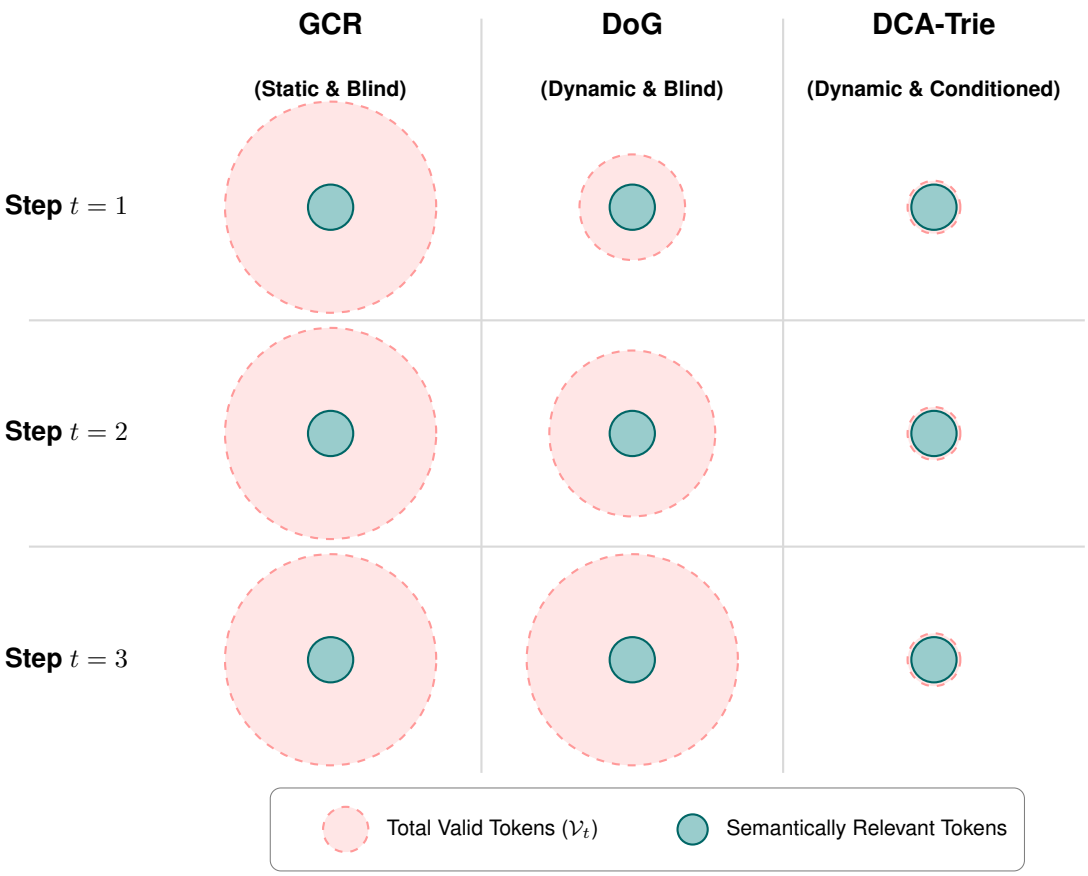

Figure 2.8 Evolution of the constraint valid set across decoding steps.

2.6 Beam Search Decoding and Prefix-Tree Constraint Integration: Mechanics, Efficiency, and the Step-Wise Constraint Problem

DCA-Trie v2 operates step-wise during beam search. Reviewers evaluating the method chapter need to understand both the beam search mechanism and how prefix-tree constraints interact with beam expansion before the DCA-Trie design can be followed. This section provides that foundation.

2.6.1 Beam Search: Approximate Global Optimization Over Autoregressive Sequences

Beam search is the standard approximate decoding algorithm for autoregressive sequence generation (Holtzman *et al.*, 2020). The algorithm maintains a *beam* (a set of k partial sequences) at each generation step. At step t, every beam hypothesis is expanded by every vocabulary token, masked logit scores are computed, and only the top-k hypotheses by cumulative log-probability are retained. Formally, for each beam hypothesis $h \in \mathcal{B}_t$:

$$\text{SCORE}(h\|v) = \text{SCORE}(h) + \log \text{SOFTMAX}\big(\tilde{\boldsymbol{\alpha}}_t\big)[v], \quad \forall\, v \in \mathcal{V} \text{ with } \tilde{\boldsymbol{\alpha}}_t[v] > 0 \tag{2.8}$$

The updated beam is $\mathcal{B}_{t+1} =$ top-k over all (h, v) pairs by $\text{SCORE}(h\|v)$. The beam width k controls the exploration-exploitation tradeoff: $k = 1$ reduces to greedy decoding; larger k explores a wider portion of the generation space at the cost of k-fold compute per step.

When combined with the logit mask from Equation (2.5), beam search operates exclusively within the valid subspace defined by the constraint oracle. The KG-Trie defines this subspace structurally; DCA-Trie's semantic filter narrows it further, reducing the number of (h, v) candidate pairs that must be scored at each beam expansion step. This reduction is what makes DCA-Trie computationally competitive with GCR despite the additional scoring overhead.

2.6.2 Prefix Trees as Constraint Oracles: Efficiency, Precomputation, and the Dynamic Rebuilding Cost

A prefix tree (trie; Fredkin 1960) over KG paths supports $O(d)$ lookup at each generation step: given the current partial sequence $y_{<t}$, the set of valid next tokens is exactly the set of children of the current trie node. This is the efficiency insight of Willard and Louf (2023): precomputing the trie amortises the constraint lookup cost across all generation steps, making constrained decoding practical at inference time. The trie construction complexity is $O(|\mathcal{P}| \times L)$, where $|\mathcal{P}|$ is the number of valid paths encoded and L is the maximum path length.

DoG's step-wise expansion sacrifices this benefit by rebuilding the trie at each step at a cost of $O(|\mathcal{N}_t| \times d)$ per step, where $|\mathcal{N}_t|$ is the number of newly reachable entities at step t. While not prohibitive per step, this cost accumulates over L steps and eliminates the precomputation advantage entirely. DCA-Trie v1 (static semantic filtering at construction time) preserves the precomputation benefit entirely, since the filtered trie is built once before generation. DCA-Trie v2 (dynamic step-wise expansion with semantic filtering) incurs per-step rebuilding overhead similar to DoG, but the semantic filter prunes most candidate expansions, so the effective expansion set $|\mathcal{N}_t^{\text{filtered}}|$ is substantially smaller than $|\mathcal{N}_t|$ in DoG.

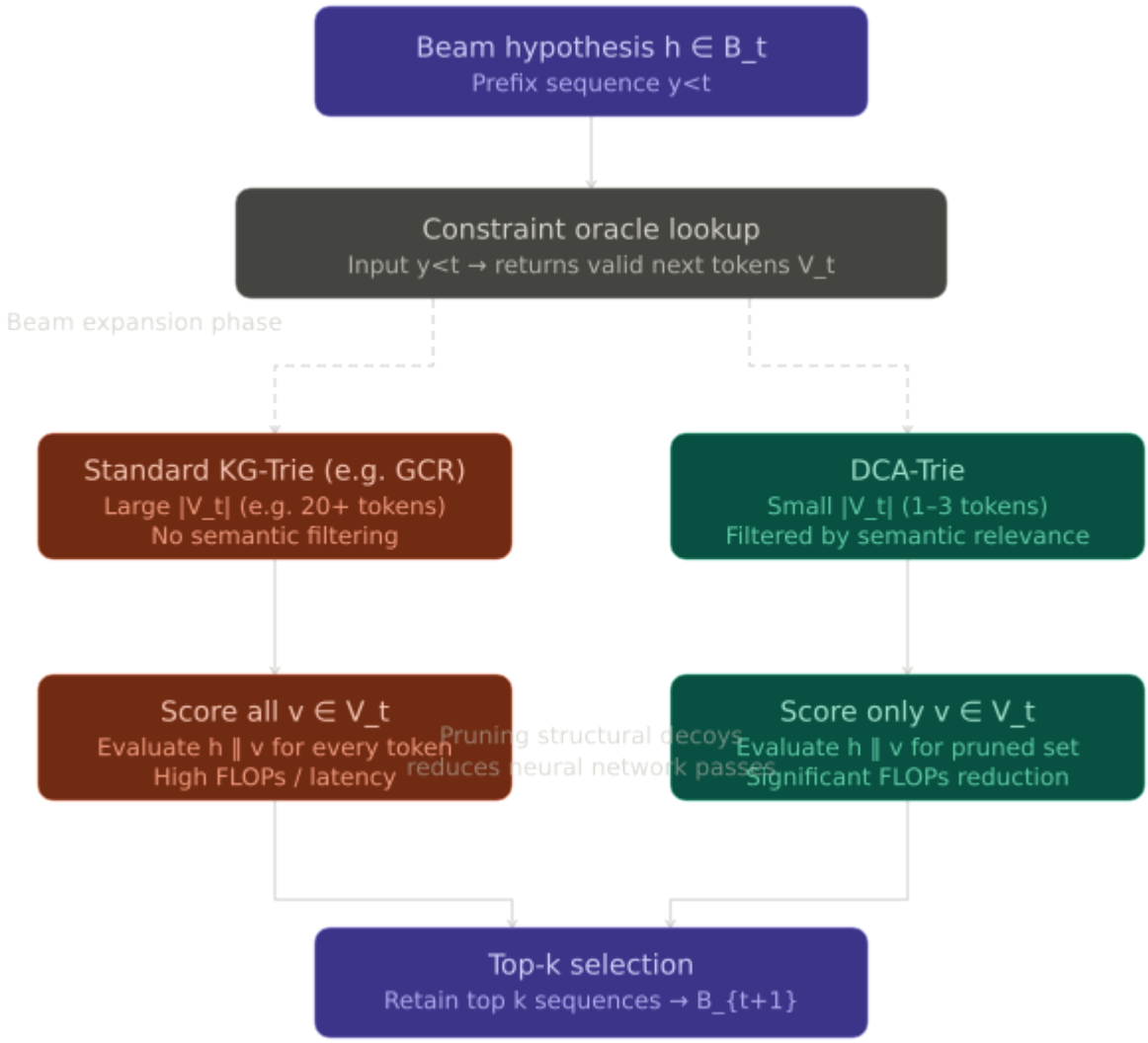

Figure 2.9 Interaction between prefix-tree constraint lookup and beam search expansion.

2.6.3 The Step-Wise Constraint Problem: When the Constraint Must Know What Has Been Generated

The step-wise constraint problem is the core design challenge motivating DCA-Trie. In GCR, the constraint oracle is computed once before generation and queried at each step without memory of what has been generated beyond the structural trie path. In DCA-Trie v2, the oracle is updated after each committed entity to reflect the semantic direction of the current reasoning chain. This requires the oracle to maintain state across generation steps; specifically, the partial generation $y_{<t}$ and the question embedding must be re-incorporated at each update.

The formal requirement is:

$$\mathcal{W}_{\text{val}}^{\text{DCA}}(t) \;=\; f\big(\mathcal{G},\, \mathcal{E}_q,\, q,\, y_{<t}\big) \tag{2.9}$$

where the function f must satisfy three conditions. (i) *Structural faithfulness*: every token $v \in \mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ corresponds to a valid KG path prefix in $\mathcal{G}$. (ii) *Semantic relevance*: the proportion of $\mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ consisting of semantically irrelevant paths is strictly less than that of $\mathcal{W}_{\text{val}}^{\text{GCR}}$. (iii) *Recall preservation*: the gold answer path is never excluded from $\mathcal{W}_{\text{val}}^{\text{DCA}}(t)$, with false negative rate held below 5% on the WebQSP validation set. These three conditions constitute the formal specification of DCA-Trie, evaluated quantitatively in Chapter 4.

2.7 Retrieval-Augmented Generation: Architectural Variants, KG-Augmented Approaches, and the Hard Limit of Soft Grounding

Retrieval-augmented generation is the dominant paradigm for grounding LLM outputs in external knowledge. This section reviews the foundational RAG architecture, KG-specific retrieval variants, and the latest advanced RAG systems, before providing a formal impossibility argument establishing that no RAG variant can achieve the structural faithfulness guarantees that constrained decoding provides.

2.7.1 The RAG Paradigm: Dense Retrieval, Fusion-in-Decoder, and the Soft Grounding Ceiling

Lewis *et al.* (2020) introduce Retrieval-Augmented Generation, in which a dense retriever selects relevant passages from an external knowledge source based on embedding similarity, and a reader LLM generates the answer conditioned on both the question and the retrieved passages. The retriever is typically implemented as a dual-encoder (Dense Passage Retrieval; Karpukhin *et al.* 2020), computing embedding similarity between the question and candidate passages offline. Fusion-in-Decoder (Izacard and Grave, 2021) extends the original RAG architecture by encoding all retrieved passages independently through the encoder and fusing their representations in the decoder, improving coverage across a larger set of retrieved documents.

The critical shared limitation of these approaches is architectural: retrieved context is a *soft influence*: the LLM is free to generate any token regardless of the retrieved material. The retriever controls what the LLM *sees*; the constraint oracle controls what the LLM may *generate*. RAG addresses the former; constrained decoding addresses the latter. Agrawal *et al.* (2024a) document the concrete consequence: even with explicit KG access, RAG-augmented models produce hallucinated reasoning steps in a substantial fraction of cases.

2.7.2 KG-Augmented Retrieval: Structured Context Without Structural Guarantees

Several works augment the RAG paradigm with KG subgraphs rather than text passages, bringing the approach closer in spirit to constrained KG decoding. He *et al.* (2021b) use attention over retrieved KG subgraphs to guide multi-hop reasoning chains. Yasunaga *et al.* (2021) combine text and KG context via graph neural networks, allowing the model to attend jointly to retrieved passages and KG neighbourhoods. Shi *et al.* (2021) retrieve KG paths as natural language context strings.

For each of these approaches, the faithfulness limitation is identical: the retrieved KG subgraph or path is provided to the LLM as additional input context. The LLM may attend to it, partially ignore it, or generate claims that go beyond it. The KG is advisory, not constraining. This is the decisive architectural distinction from GCR and DCA-Trie, where the KG is *mechanically* enforced through the logit mask rather than provided as contextual suggestion.

2.7.3 Advanced RAG Variants: Self-RAG, FLARE, IRCoT, and GraphRAG

Four advanced RAG variants warrant individual analysis as potential alternatives to constrained decoding that a reviewer may raise.

Self-RAG (Asai *et al.*, 2024) augments the standard RAG pipeline by inserting special reflection tokens that allow the model to decide dynamically when retrieval is needed and whether retrieved passages are relevant or supportive. This improves retrieval precision and citation accuracy. However, Self-RAG does not modify the token selection mechanism: the LLM remains free to generate any token at any step, and the reflection tokens are themselves generated autoregressively. Unlike DCA-Trie, Self-RAG cannot guarantee structural faithfulness.

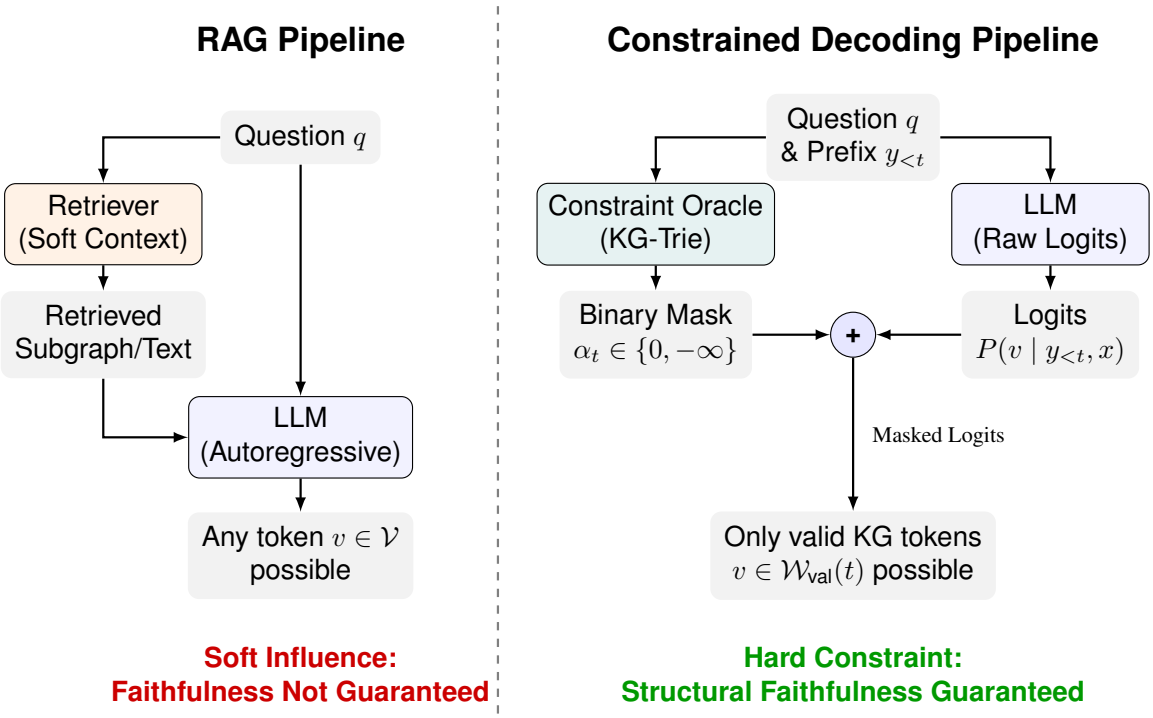

Figure 2.10 Architectural comparison of standard Retrieval-Augmented Generation (left) and KG-constrained decoding (right).

FLARE (Jiang *et al.*, 2023) uses the LLM's next-token prediction probability as a signal to trigger retrieval when confidence falls below a threshold. This addresses the question of *when* to retrieve but leaves the generation faithfulness problem untouched: retrieved context is again a soft input influence.

IRCoT (Trivedi *et al.*, 2023) interleaves retrieval with chain-of-thought reasoning steps, performing a retrieval call at each CoT step. This substantially improves multi-hop accuracy by ensuring that each reasoning step is informed by freshly retrieved context. Each CoT step nevertheless remains an unconstrained autoregressive generation, retaining strictly positive probability over all vocabulary tokens.

GraphRAG (Edge *et al.*, 2024) is the strongest RAG-based competitor to DCA-Trie, as it explicitly retrieves community-structured graph context to support complex relational queries. GraphRAG retrieves graph context but does not constrain decoding: a generation conditioned on GraphRAG context can still produce tokens that correspond to no KG triplet in $\mathcal{T}$.

For every variant, the same single sentence applies: unlike DCA-Trie, none of these methods modifies the token selection mechanism and therefore none can guarantee structural faithfulness.

2.7.4 The Formal Impossibility of Structural Faithfulness Under Soft Grounding

The following argument establishes that no retrieval-augmented approach can provide structural faithfulness guarantees as defined in this dissertation.

Claim: Any approach that grounds LLM generation by modifying its input (retrieved context, subgraphs, passages) without modifying the logit computation mechanism cannot guarantee structural faithfulness with probability one.

Proof sketch: Structural faithfulness requires that every generated token y_t corresponds to a verified KG triplet. This requires that every token $v \notin \mathcal{W}_{\text{val}}(t)$ has exactly zero probability at step t, i.e., $P(v \mid y_{<t}, x) = 0$ for all invalid v. Under any RAG approach, the LLM's raw logit distribution assigns strictly positive logit to all tokens in $\mathcal{V}$ (or at least to all tokens not explicitly set to $-\infty$). Retrieval augmentation modifies the input x to include retrieved context $\hat{x}$, but does not modify the logit computation $P(\cdot \mid y_{<t}, \hat{x})$: the distribution remains over the full vocabulary with strictly positive probabilities. Therefore, for any RAG approach, there exists strictly positive probability of generating a non-KG-valid token at any step. Probability-one structural faithfulness is not achievable under this architecture. The only mechanism that achieves it is a constraint oracle that explicitly sets $\alpha_t[v] = -\infty$ for all invalid v: the logit-masking mechanism of Equation (2.5), as implemented in GCR, DoG, and DCA-Trie. Luo *et al.* (2024) provide empirical confirmation: GCR achieves 100% structural faithfulness where RAG-based approaches do not. □

2.8 Semantic Relevance Scoring with Sentence Transformer Embeddings: Training Objectives, Architecture Selection, and Application to Constraint Oracle Construction

This section justifies every design choice in DCA-Trie's semantic relevance scorer, responding in advance to the reviewer objections: "Why not BM25?", "Why not use the LLM's own hidden states?", and "Why all-MiniLM-L6-v2 over a stronger encoder?"

2.8.1 Sentence-BERT: Training Objective, Architecture, and the Semantic Similarity Property

Reimers and Gurevych (2019) introduce Sentence-BERT (SBERT), a modification of the BERT architecture trained specifically to produce dense vector representations in which

semantically similar sentences have high cosine similarity. The architecture uses a BERT-like transformer encoder with mean-pooling over token representations to produce a fixed-length sentence embedding. SBERT is trained using a Siamese network on labelled sentence pairs. The classification-objective training loss for a pair (s_1, s_2) with embeddings $\mathbf{u} = E(s_1)$ and $\mathbf{v} = E(s_2)$ is:

$$\mathcal{L} = \textsc{CrossEntropy}\Big(\text{softmax}\big(\mathbf{W}_t \cdot (\mathbf{u},\ \mathbf{v},\ |\mathbf{u} - \mathbf{v}|)\big),\ y\Big) \tag{2.10}$$

where $(\mathbf{u}, \mathbf{v}, |\mathbf{u}-\mathbf{v}|)$ is the concatenation of the two embeddings with their element-wise absolute difference, $\mathbf{W}_t$ is a trainable classification head, and $y \in \{\text{entailment}, \text{neutral}, \text{contradiction}\}$. This objective pulls embeddings of entailed pairs together and pushes contradiction pairs apart in the representation space. At inference time, $\cos(E(s_1), E(s_2))$ measures the degree of semantic alignment between two sentences without requiring the classification head, enabling cosine similarity to serve as a reliable semantic relevance proxy without any task-specific fine-tuning.

2.8.2 Encoder Architecture Selection: all-MiniLM-L6-v2 vs Alternatives

DCA-Trie requires a semantic encoder that is invoked at every beam search step during constrained decoding. The selection of the encoder architecture therefore involves a tradeoff between semantic quality and inference latency that is more stringent than standard offline retrieval settings. Table 2.2 presents a systematic comparison of five candidate architectures.

Table 2.2 Comparison of sentence encoder architectures.

Encoder	**Params**	**Emb. Dim**	**Latency (ms)**	**STS-B**	**Step-Wise?**
all-MiniLM-L6-v2 *(selected)*	22M	384	∼80	82.3	Yes
all-mpnet-base-v2	110M	768	∼320	84.6	No
BGE-M3	568M	1024	>500	87.0	No
LLM hidden states	—	varies	high	—	No
BM25 (lexical)	—	—	<5	63.0	Yes

Latency figures are approximate per-query values on a single CPU core. STS-B scores are Spearman correlations reported in the sentence-transformers library benchmarks (Reimers and Gurevych, 2019).

all-MiniLM-L6-v2 (Reimers and Gurevych, 2019) is a six-layer MiniLM model distilled from a larger teacher, producing 384-dimensional embeddings with approximately 22M parameters and roughly 80ms inference latency on CPU. It achieves strong semantic performance on STS benchmarks while maintaining the lowest inference latency among semantically competitive models.

all-mpnet-base-v2 achieves stronger semantic performance but requires approximately 110M parameters and incurs around 4× the inference latency. For iterative beam-search decoding in which the scorer is called at every step, this latency accumulates unacceptably over the full generation length.

BGE-M3 (Chen *et al.*, 2024) is a state-of-the-art multilingual retrieval encoder with stronger performance than MiniLM, but its 568M parameters and resulting inference latency make it infeasible for step-wise decoding on a single A100 GPU.

LLM hidden states as a relevance scorer would be circular: the LLM's last-layer representations encode the same parametric biases that cause hallucination. Using them as a relevance filter does not introduce any external grounding and defeats the purpose of the constraint mechanism.

BM25 is fast and interpretable but a lexical retrieval method: it cannot capture semantic similarity between paraphrase-equivalent KG paths that share the same meaning but no surface tokens, which is common in Freebase relation strings.

The conclusion is that all-MiniLM-L6-v2 provides the optimal balance of semantic quality and inference speed for iterative step-wise constraint filtering.

2.8.3 Cosine Similarity as a Hard Constraint Signal: From Soft Retrieval to Binary Admission

In standard dense retrieval, cosine similarity produces a continuous relevance ranking over candidate documents. DCA-Trie's key conceptual contribution in this component is the conversion of this continuous signal into a *binary admission criterion* that determines which tokens the LLM is permitted to generate.

Given a candidate KG path p expressed as a natural language string $(e_0\, r_1\, e_1\, r_2\, e_2 \ldots)$, the question q, and the partial generation $y_{<t}$, the semantic relevance score is:

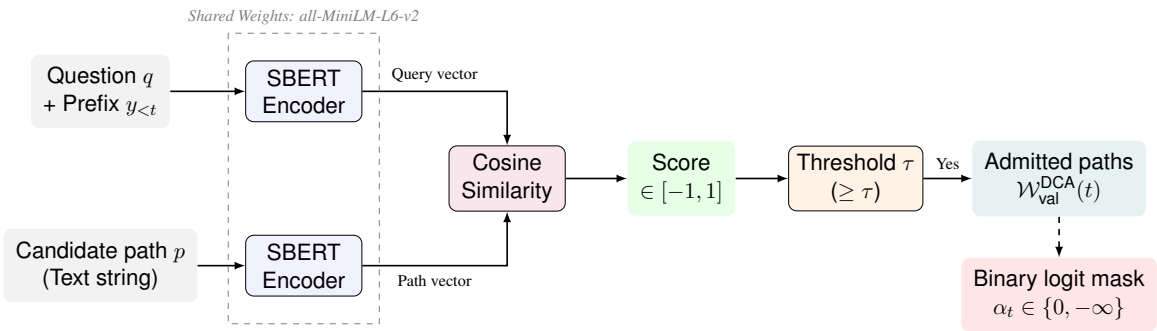

Figure 2.11 End-to-end semantic scoring and binary admission pipeline.

$$\text{SCORE}(p,\ q,\ y_{<t}) \ = \ \cos\big(E(p),\ \ E(\text{concat}(q,\ y_{<t}))\big) \tag{2.11}$$

where $E(\cdot)$ is the all-MiniLM-L6-v2 encoder. The binary admission criterion is: admit p to $\mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ if and only if $\text{SCORE}(p, q, y_{<t}) \geq \tau$. This threshold function converts the continuous cosine similarity into a binary mask entry, integrating semantic relevance directly into the logit-masking framework of Equation (2.5). The threshold τ is the key hyperparameter controlling the precision-recall tradeoff of the relevance filter; its selection procedure is described in Chapter 3 and evaluated in Chapter 4.

2.9 Related Work

Sections 2.2–2.8 have surveyed the landscape thematically. This section positions DCA-Trie against the specific papers a reviewer is most likely to cite as prior art.

2.9.1 KG-Constrained Decoding: GCR, DoG, and the Unresolved Permissiveness Problem

GCR (Luo *et al.*, 2024) constructs a static KG-Trie from $f(\mathcal{G}, \mathcal{E}_q)$ before generation, achieving 92.6 Hits@1 on WebQSP and 100% structural faithfulness; however, the valid token set is semantically blind and computationally identical at every generation step (Section 2.5). DoG (Li *et al.*, 2024) extends GCR with step-wise topological trie expansion after each committed entity, improving dynamic responsiveness at the cost of per-step reconstruction; but as shown in Equation (2.7), the expansion rule admits all KG neighbours regardless of semantic relevance, so the permissiveness problem persists. Both works leave unresolved the central limitation: the valid token set contains thousands of structurally correct but contextually irrelevant paths at every generation step, forcing the LLM to rely on hallucination-prone

parametric knowledge for semantic discrimination. DCA-Trie addresses this directly by conditioning the valid token set on the input question semantics and the partial generation, formally upgrading the oracle from $f(\mathcal{G}, \mathcal{E}_q)$ to $f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$ (Equation 2.9). DCA-Trie is designed as a direct extension of GCR and DoG, preserving their structural faithfulness guarantee while resolving the permissiveness problem neither addresses.

Table 2.3 Primary related work comparison.

Method	Oracle	Faithfulness	Sem. Cond.	Dynamic	Precomp.	Hits@1
GCR (Luo *et al.*, 2024)	$f(\mathcal{G}, \mathcal{E}_q)$	100%	No	No	Yes	92.6
DoG (Li *et al.*, 2024)	$f(\mathcal{G}, \mathcal{E}_q, t)$	**100%**	**No**	**Yes**	**No**	**90.99** (WebQSP) / **76.08** (CWQ)
RouterKGQA (Yuan *et al.*, 2026)	Routing + filter	Partial	Yes (answer)	Yes	N/A	+0.49 vs GCR
DCA-Trie v1 (proposed)	$f(\mathcal{G}, \mathcal{E}_q, q)$	**100%**	**Yes (token)**	**No**	**Yes**	**TBD**
DCA-Trie v2 (proposed)	$f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$	**100%**	**Yes (token)**	**Yes**	**Partial**	**TBD**

2.9.2 Hallucination Mitigation and KGQA: The Soft-to-Hard Spectrum

Soft grounding approaches constitute one cluster of related work. RAG (Lewis *et al.*, 2020), Self-RAG (Asai *et al.*, 2024), GraphRAG (Edge *et al.*, 2024), DAGQA (He *et al.*, 2021b), QA-GNN (Yasunaga *et al.*, 2021), and self-consistency (Wang *et al.*, 2023) all improve faithfulness in expectation by providing the LLM with relevant external context or by aggregating over multiple generation samples. As established formally in Section 2.7, none of these approaches modifies the token selection mechanism: they cannot guarantee structural faithfulness with probability one, because every token in $\mathcal{V}$ retains strictly positive probability at every step.

Format-constrained decoding constitutes the second cluster. GCD (Geng *et al.*, 2023), Outlines (Willard and Louf, 2023), and entity-constrained generation (De Cao *et al.*, 2021) establish the logit-masking paradigm on which DCA-Trie builds. These works constrain output format or entity vocabulary rather than multi-hop KG path faithfulness: their constraint oracle is a function of a grammar or entity list, not of a knowledge graph with relational structure. DCA-Trie inherits the logit-masking mechanism from this line of work and extends it to semantically conditioned KG path constraints.

2.9.3 Semantic Similarity for Graph Reasoning: The Retrieval-Constraint Boundary

Several works apply semantic similarity signals in graph-based reasoning. Path retrieval for KGQA (Shi *et al.*, 2021; Saxena *et al.*, 2022) uses semantic similarity to rank candidate

paths for retrieval. Attention-guided graph traversal in DAGQA (He *et al.*, 2021b) uses attention weights to focus multi-hop reasoning.Semantic path reranking approaches select among candidate answers after generation. In every case, semantic similarity operates as a *soft signal* that influences which paths the LLM sees or which answer is selected after generation; it does not apply semantic similarity as a hard binary admission criterion that determines which tokens the LLM is *permitted* to generate.

This distinction between soft signals and hard constraints defines the exact novelty boundary of DCA-Trie. No prior work combines semantic relevance scoring with hard constraint oracle construction. Works that provide structural faithfulness (GCR,DoG) do not condition on semantics. Works that use semantic similarity (path retrieval, reranking) do not provide structural faithfulness. DCA-Trie is the first framework to achieve both simultaneously, and the Semantic Irrelevance Ratio introduced in Chapter 3 is the first metric designed to measure this property directly.

2.9.4 Concurrent and Subsequent Work: Positioning DCA-Trie

The research landscape has evolved rapidly since the foundational works of GCR (Luo *et al.*, 2024) and DoG (Li *et al.*, 2024). Most notably, **RouterKGQA** (Yuan *et al.*, 2026) proposes a specialized–general model routing framework that achieves strong performance with minimal inference cost. RouterKGQA employs constraint-aware reasoning paths and uses SBERT (nomic-embed-text-v1) for semantic filtering during answer selection, achieving 3.57 F1 point improvements over GCR on WebQSP and CWQ while requiring only 1.15 LLM calls per question (Yuan *et al.*, 2026).

Differentiation of DCA-Trie

While RouterKGQA and DCA-Trie both employ semantic relevance scoring, they operate at fundamentally different architectural levels:

Table 2.4 Architectural comparison of RouterKGQA and DCA-Trie.

Aspect	RouterKGQA	DCA-Trie
Intervention Point	Answer filtering (post-generation)	Token constraint (during generation)
Mechanism	Routes between specialized and general models	Modifies the constraint oracle directly
Semantic Scoring	Filters final candidate answers	Filters valid token sets at each decoding step
Faithfulness Guarantee	Probabilistic (repair when needed)	Deterministic (hard constraint at every step)
Metric Innovation	Efficiency (LLM calls)	Constraint quality (SIR)

RouterKGQA's semantic filtering operates on complete reasoning paths after generation, whereas DCA-Trie's semantic scoring constrains the *decoding process itself*–ensuring that the LLM never sees semantically irrelevant tokens during generation. This distinction is critical: DCA-Trie reduces the cognitive load on the LLM by narrowing the valid vocabulary at each step, while RouterKGQA relies on the LLM to generate paths then filters retrospectively.

Furthermore, DCA-Trie introduces the **Semantic Irrelevance Ratio (SIR)**, the first metric to isolate and quantify constraint permissiveness independently of downstream generation accuracy. RouterKGQA does not provide a direct measure of how many structurally valid but semantically irrelevant paths enter its candidate set; DCA-Trie's SIR metric fills this evaluation gap.

2.10 Synthesis of the Literature: Three Unresolved Gaps and the DCA-Trie Contribution

This section synthesizes the findings of the literature review by formalizing three distinct research gaps. Each limitation is then explicitly mapped to a proposed component of the DCA-Trie architecture and its corresponding project objective, establishing the foundation for the methodology detailed in Chapter 3.

2.10.1 Three Unresolved Gaps at the Intersection of Structural Faithfulness, Semantic Relevance, and Constraint Measurability

Gap 1: The Permissiveness Problem. GCR and DoG construct constraint oracles as $f(\mathcal{G}, \mathcal{E}_q)$, independent of question semantics and partial generation. For a three-hop CWQ question, Equation (2.4) shows this produces a valid token set containing up to 24,000 structurally correct but semantically irrelevant paths, forcing the LLM to use hallucination-prone

parametric knowledge for discrimination, precisely the mechanism constrained decoding was designed to suppress (Luo *et al.*, 2024; Li *et al.*, 2024).

Gap 2: The Static-Dynamic Efficiency Tradeoff. GCR's precomputed static trie is computationally efficient but semantically blind. DoG's step-wise topological expansion introduces dynamism at the cost of per-step trie reconstruction and without semantic filtering (Li *et al.*, 2024; Willard and Louf, 2023). No existing framework achieves semantic responsiveness while preserving the efficiency of a precomputed oracle.

Gap 3: The Absence of a Permissiveness Metric. No existing work quantifies constraint permissiveness independently of downstream answer accuracy, making it impossible to measure progress on reducing irrelevant paths in the valid set directly or to compare frameworks on this property (Luo *et al.*, 2024; Li *et al.*, 2024).

Table 2.5 Consolidated gap matrix for KG-constrained decoding frameworks.

Framework	**Faithfulness**	**Sem. Cond.**	**Dynamic**	**Precomp.**	**SIR**
GCR (Luo *et al.*, 2024)	✓	No	No	✓	No
DoG (Li *et al.*, 2024)	✓	No	✓	No	No
RouterKGQA (Yuan *et al.*, 2026)	∼[a]	✓(answer)	✓	N/A	No
DCA-Trie v1 (proposed)	✓	✓(token)	No	✓	✓
DCA-Trie v2 (proposed)	✓	✓(token)	✓	Partial	✓

[a] Probabilistic faithfulness via repair mechanism; not guaranteed.

2.10.2 DCA-Trie as the Unified Response: Gap-to-Objective Mapping

Each gap maps directly to a DCA-Trie component and project objective. **Gap 1** is addressed by DCA-Trie's semantic relevance scorer applied as a hard admission criterion, formally upgrading the constraint oracle from $f(\mathcal{G}, \mathcal{E}_q)$ to $f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$, targeting project Objectives ii, iii, and iv. **Gap 2** is addressed by the two-variant design: DCA-Trie v1 applies semantic filtering at trie construction time, preserving precomputation efficiency (Objective iii); DCA-Trie v2 applies step-wise dynamic expansion with semantic filtering for maximum contextual responsiveness (Objective iv). **Gap 3** is addressed by the Semantic Irrelevance Ratio (SIR), a new metric introduced in Chapter 3 that quantifies the proportion of the valid set consisting of semantically irrelevant paths (Objective i).

DCA-Trie is the first constrained decoding framework to simultaneously address all three gaps, providing structural faithfulness, semantic relevance conditioning, and a dedicated permissiveness metric. RouterKGQA addresses efficiency through routing but does not provide deterministic faithfulness guarantees or constraint-level evaluation; DCA-Trie fills these critical gaps. These constitute the three novel contributions of this dissertation.

Table 2.6 DCA-Trie contribution mapping.

Research Gap	**DCA-Trie Component**	**Objective**	**Evaluated In**
Gap 1: Permissiveness Problem	Semantic scorer + threshold τ	Obj. ii, iii, iv	Ch. 3–4
Gap 2: Static-Dynamic Tradeoff	v1 (static) and v2 (dynamic) variants	Obj. iii, iv	Ch. 3–4
Gap 3: No Permissiveness Metric	Semantic Irrelevance Ratio (SIR)	Obj. i	Ch. 3–4

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter establishes the core methodology of the DCA-Trie framework, extending from formal problem specification and architectural algorithmic design through to the evaluation protocol. The chapter is organised as follows. Section 3.2 formalises the research problem, deriving three strict constraints that any capable solution must satisfy. Section 3.3 presents the primary research questions structuring the experimental logic. Section 3.4 introduces the Semantic Irrelevance Ratio (SIR), a novel metric designed to continuously quantify constraint permissiveness independently of downstream generation accuracy. Section 3.5 specifies the semantic relevance scoring mechanism functioning as the foundational computational filter of DCA-Trie. Sections 3.6 and 3.7 detail the design, algorithmic implementation, complexity bounds, and faithfulness guarantees of DCA-Trie v1 and v2, respectively. Section 3.8 outlines the baseline systems against which the framework is evaluated, while Section 3.9 lays out the empirical protocol, datasets, and hyperparameters. Section 3.10 specifies targeted ablation and sensitivity analyses to isolate the contribution of critical hyperparameters. Finally, Section 3.11 acknowledges specific scope boundaries and anticipated limitations that bound the empirical assertions in Chapter 4.

3.2 Formal Problem Specification

3.2.1 Restating the Core Limitation of Existing Frameworks

As elucidated in Chapter 2, existing knowledge graph-constrained generation frameworks, prominently GCR (Luo *et al.*, 2024) and DoG (Li *et al.*, 2024), instantiate their constraint oracles as deterministic functions evaluated over the knowledge graph $\mathcal{G}$ and the identified question entities $\mathcal{E}_q$, expressed formally as $f(\mathcal{G}, \mathcal{E}_q)$. Whether computed entirely a priori or expanded topologically during decoding, the mathematical consequence is that the permitted valid token set $\mathcal{W}_{\text{val}}(t)$ at any arbitrary decoding step t resolves to:

$$\mathcal{W}_{\text{val}}^{\text{GCR}}(t) = f(\mathcal{G},\, \mathcal{E}_q) \quad \forall\, t \tag{3.1}$$

Crucially, this specification is entirely independent of two critical contextual vectors: the semantic intent embedded within the natural language question q, and the dynamically evolving reasoning trajectory captured by the partial generation sequence $y_{<t}$. Consequently, all structurally accessible knowledge graph paths manifest symmetrically within the constraint hypothesis space, irrespective of their logical utility or factual relevance to the target query. For complex, multi-hop queries executing over dense subgraphs in Freebase, the combinatorial upper bound on this admitted valid set evaluates strictly to $|\mathcal{E}_q| \times d^3 \approx 8{,}000$ distinct traversable paths (referenced from Equation 2.4 in Chapter 2). Of this mathematically profound subset, strictly one path maps to the verifiable gold answer.

Such architectural paradigms erroneously conflate two orthogonal computational requirements. *Structural validity* demands that any token permitted by the constraint oracle corresponds deterministically to a topological edge within the underlying knowledge schema—a baseline criteria satisfied universally by modern approaches. Conversely, *Contextual relevance* necessitates that the permitted parameter space accurately reflects the task-specific semantic vector dictating generation step t—a stringent requirement universally ignored by contemporary mechanisms. This chasm between rigid topological correctness and fluid semantic applicability encapsulates the formal definition of the *permissiveness problem*.

3.2.2 The Upgraded Oracle Specification

The DCA-Trie architecture resolves the permissiveness problem by formalising an upgraded mathematical constraint oracle that unifies structural accessibility with semantic integration. This upgraded function dynamically absorbs both the input semantic tensor and the unfolding generation prefix:

$$\mathcal{W}_{\text{val}}^{\text{DCA}}(t) = f(\mathcal{G},\, \mathcal{E}_q,\, q,\, y_{<t}) \tag{3.2}$$

where q encapsulates the natural language query structure and $y_{<t} = (y_1, \ldots, y_{t-1})$ signifies the precise token history resolved at step t. To assert methodological completeness, this upgraded constraint vector must simultaneously satisfy three absolute conditions, which precisely operationalise the central objectives detailed in Chapter 1:

1. **Structural Faithfulness (Condition F):** Every candidate token $v \in \mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ at every generation step t must map conclusively to a mathematically valid sequence prefix of an existing path $p \in \mathcal{G}$. This axiom ensures that DCA-Trie rigorously observes a 100%

structural faithfulness baseline comparable to GCR and DoG, definitively preempting the introduction of any ungrounded lexical hallucinations into the constrained valid set.

2. **Semantic Relevance Reduction (Condition R):** The computed Semantic Irrelevance Ratio (SIR) of $\mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ must demonstrate a strict mathematical reduction compared to the equivalent metric over $\mathcal{W}_{\text{val}}^{\text{GCR}}(t)$ when averaged globally across statistical evaluation corpora. Formally, denoting SIR as the primary diagnostic specified in Section 3.4:

$$\overline{\text{SIR}}\big(\mathcal{W}^{\text{DCA}}\big) < \overline{\text{SIR}}\big(\mathcal{W}^{\text{GCR}}\big) \tag{3.3}$$

3. **Recall Preservation (Condition P):** Evaluated within the context of semantic pruning, the verifiable gold reference path must mathematically evade spurious elimination by the scoring filter. Operationally, the empirical false negative rate (FNR) on the constrained WebQSP validation locus must evaluate strictly below the 5% threshold:

$$\text{FNR} = \frac{|\{q \in \mathcal{Q}_{\text{val}} : p_q^* \notin \mathcal{W}_{\text{val}}^{\text{DCA}}(t)\}|}{|\mathcal{Q}_{\text{val}}|} < 0.05 \tag{3.4}$$

where p_q^* resolves to the gold truth mechanism for question q and $\mathcal{Q}_{\text{val}}$ encapsulates the dedicated 100-question held-out validation segment.

These three fundamental stipulations transcend abstract diagnostic desiderata; they exist as deterministic targets quantified exhaustively in the Chapter 4 empirical framework. Condition F is structurally verified by the raw faithfulness output metric. Condition R translates directly to empirical observation via the proposed Semantic Irrelevance Ratio (SIR). Condition P evaluates quantitatively against precision thresholds. Operating collectively, they form the core mathematical scaffolding constraining the DCA-Trie implementation.

3.3 Research Questions

The empirical validation strategy executed in Chapter 4 is rigidly structured around four explicit research questions. These queries strictly operationalise the foundational project objectives delineated in Chapter 1 and establish a formal methodology for assessing the viability of the DCA-Trie architecture.

RQ1 (Permissiveness Baseline Characterisation): To what extent does the constraint permissiveness problem degrade structural efficiency within GCR's static KG-Trie across the WebQSP and CWQ corpora, and how does this degradation scale mathematically as a function of question hop-depth? This inquiry rigorously operationalises Objective i, strictly establishing the quantitative failure baseline against which DCA-Trie's strategic architectural interventions will be systematically evaluated.

RQ2 (Target Truth Accuracy): Do the dynamic architectural permutations, namely DCA-Trie v1 and v2, fundamentally match or strictly surpass the generative baseline accuracy (measured via primary Hits@1 and F1) established by GCR across the specified multi-hop QA benchmarks? This parameter isolates the foundational query: whether aggressive semantic pruning intrinsically elevates terminal answer fidelity by enforcing dense reasoning vectors, or inadvertently deteriorates global performance by masking legitimate graph routes.

RQ3 (Strict Structural Faithfulness Conservation): Does the integration of aggressive mathematical relevance gating explicitly guarantee the 100% baseline structural faithfulness boundary inherent to the underlying models? This question definitively verifies Condition F, mathematically asserting that the integrated similarity semantic scorer strictly intercepts no procedurally invalid or structurally ungrounded hallucinated tokens into the terminal output space.

RQ4 (Semantic Irrelevance Mitigation Efficiency): What strict geometric reduction in the Semantic Irrelevance Ratio (SIR) do the DCA-Trie architectures introduce relative to the unconstrained GCR baseline, and does this mathematical mitigation coefficient proportionally escalate with topological traversal hop-depth relative to Equation (3.3)? This query conclusively determines progress concerning Objective ii, presenting terminal empirical veracity validating the systematic resolution of the underlying constraint permissiveness paradox.

3.4 The Semantic Irrelevance Ratio: Definition and Measurement

3.4.1 Motivation

Standard evaluation metrics deployed for knowledge graph-constrained decoding architectures, specifically Hits@1, F1, and structural faithfulness rate, uniformly assess downstream generation outcomes: whether the terminal entity matches the gold answer, and whether the

lexical output is strictly graph-valid. However, none mathematically verify the inherent precision of the constraint mechanism itself. A decoding framework could hypothetically achieve flawless target metrics while actively propagating the permissiveness problem: the underlying large language model might arrive at the correct terminal node by successfully traversing and discriminating among thousands of structurally valid but contextually irrelevant paths relying strictly on parametric memorisation, rather than because the constraint mechanism actually narrowed the conceptual search horizon. To resolve this critical diagnostic gap, this research introduces the Semantic Irrelevance Ratio (SIR) as a deterministic mathematical quantification of constraint efficiency independent of generation fidelity.

3.4.2 Formal Definition

Let $\mathcal{P}(q, t)$ formally denote the complete set of knowledge graph paths structurally admitted by the active constraint oracle during decoding execution step t for a designated query q. A specific topological path $p \in \mathcal{P}(q, t)$ is definitively designated as *semantically irrelevant* if its vectorised cosine similarity relative to the concatenated question-context embedding evaluates below a rigidly established baseline threshold τ_{ref}:

$$\text{irrelevant}(p, q, y_{<t}) = \mathbf{1}\left[\cos\big(E(p),\ E(\text{concat}(q, y_{<t}))\big) < \tau_{\text{ref}}\right] \tag{3.5}$$

where $E(\cdot)$ represents the selected high-density `all-MiniLM-L6-v2` encoder embedding projection, and the boundary condition $\tau_{\text{ref}} = 0.3$ serves as a non-tunable static reference threshold formulated to securely delineate off-topic structural traversals without conflating underlying relevance verification with DCA-Trie's structurally dynamic admission boundary parameter τ.

Consequently, the Semantic Irrelevance Ratio for a singular evaluated question q isolating step t is strictly defined as:

$$\text{SIR}(q, t) = \frac{\sum_{p \in \mathcal{P}(q,t)} \text{irrelevant}(p, q, y_{<t})}{|\mathcal{P}(q, t)|} \tag{3.6}$$

The aggregate diagnostic SIR quantifying an entire generational framework across an independent evaluation corpus $\mathcal{Q}$ is derived by averaging uniformly over the sum of all queries

intersecting all valid generation intervals up to the deterministic maximum traversal depth L:

$$\overline{\text{SIR}} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{L_q} \sum_{t=1}^{L_q} \text{SIR}(q, t) \tag{3.7}$$

where L_q strictly dictates the cumulative hop dimension of the generated lexical reasoning chain regarding query q.

3.4.3 Properties and Interpretation

The aggregate ratio $\overline{\text{SIR}}$ mathematically inhabits the interval $[0, 1]$. A terminal limit converging on $\overline{\text{SIR}} = 1$ confirms structural catastrophe constraint-wise: every route permitted by the structural oracle inherently lacks textual relevance regarding the primary question, effectively forcing the parametric LLM layer to shoulder the absolute burden of logical traversal. By definitive contrast, an evaluated baseline of $\overline{\text{SIR}} = 0$ perfectly asserts that every generated constraint subset represents intense, contextual target relevance—affirming perfect spatial restriction. Extant frameworks including GCR and DoG systematically execute near the maximum boundary, mathematically confirming massive structural SIR overhead specifically inflating at three and four-hop traversal bounds, where exhaustive path enumeration explodes. The implementation of the DCA-Trie architecture mathematically executes an intentional, quantifiable suppression of the overall SIR calculation, aggressively bypassing semantically tangential routes leveraging the active threshold mechanisms mathematically specified via Equation (3.2). Following standardized practices, empirical SIR outputs will inherently evaluate stratified across varied discrete hop depths to determine if active semantic exclusion consistently cascades scaling parallel to mounting query complexity.

3.5 Semantic Relevance Scoring Mechanism

3.5.1 Scorer Architecture

The semantic relevance scoring module natively operationalises the foundational architectural paradigm shift from the static function $f(\mathcal{G}, \mathcal{E}_q)$ to the dynamic formulation $f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$. The mathematical operation dictates receiving an individual discrete structural candidate knowledge graph path p, the raw natural language query sequence q, and the continuous

partial derivation $y_{<t}$, synthesising these components into an executable continuous scalar relevance output within the $[0, 1]$ interval space.

An individual topological search path $p = (e_0, r_1, e_1, \ldots, r_L, e_L)$ is sequentially serialised into an interpretable natural language sequence via string concatenation bounded by space delimiters isolating consecutive structural nodes and edges:

$$\text{str}(p) = e_0 \parallel r_1 \parallel e_1 \parallel \cdots \parallel r_L \parallel e_L \tag{3.8}$$

where extracted entity descriptors mathematically map to resolved Freebase human-readable lexical names (decoded computationally from internal MIDs wherever semantically viable), and intermediate formal relation vectors denote Freebase relation schema schemas where internal programmatic underscores undergo regular expression space replacement. Converging symmetrically, the dynamic querying context synthesises string integration parsing both spatial origin and topological position separating by an explicit lexical breakpoint string:

$$\text{query}(q, y_{<t}) = q \parallel \texttt{[SEP]} \parallel \text{str}(y_{<t}) \tag{3.9}$$

where $\text{str}(y_{<t})$ effectively materialises the locally detokenised historical chain progression.

Sequentially, both string parameters $\text{str}(p)$ and $\text{query}(q, y_{<t})$ map deterministically through the robustly dense `all-MiniLM-L6-v2` transformer encoder block (Reimers and Gurevych, 2019) translating natural linguistics into unified 384-dimensional spatially dense representation coordinates. The terminal absolute relevance scalar quantifies their continuous spatial cosine geometry:

$$\text{score}(p,\ q,\ y_{<t}) = \cos\Big(E(\text{str}(p))\,,\ \ E(\text{query}(q, y_{<t}))\Big) \tag{3.10}$$

Thus, an iteratively assessed semantic candidate path p earns mathematical admission to the local continuous valid lexicon $\mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ explicitly if and only if its geometric divergence confirms $\text{score}(p, q, y_{<t}) \geq \tau$, where τ constitutes the active parameterised admission magnitude threshold. This decisive binary activation directly embeds the continuous multi-dimensional similarity computation deeply within the robust logit-masking schema originally modeled natively as Equation (2.5) in Chapter 2. The total topological execution algorithm visually

structures itself accordingly in Figure 3.1.

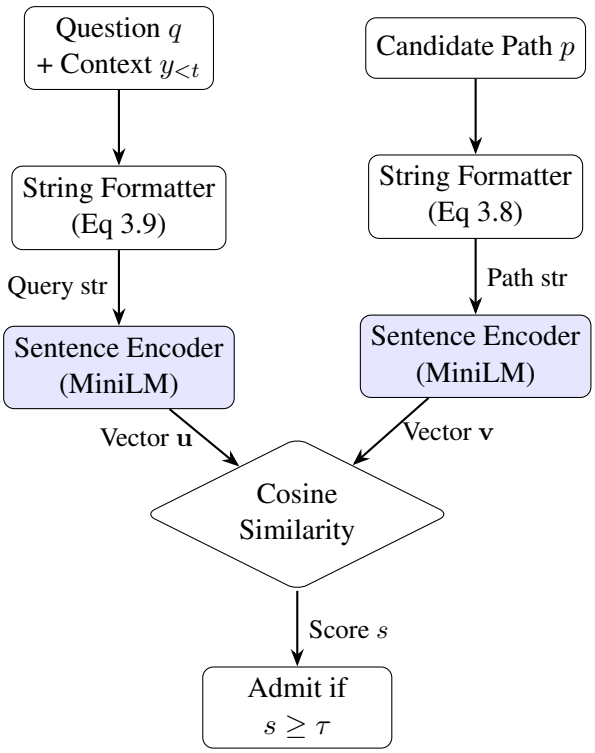

Figure 3.1 The semantic relevance scoring mechanism. Candidate knowledge graph paths and the dynamically updating query-context prefix sequence are independently routed down continuous pathways to an aggregated dense mapping defined by a tuned state-of-the-art encoder. Deterministic binary admission implements an aggressive exclusion gate parameterized upon a designated τ limit bounding DCA-Trie's contextual relevance scope.

3.5.2 Threshold Selection

The absolute admission threshold magnitude τ is systematically isolated via a precise mathematical evaluation over a structurally detached validation partition housing strictly 100 uniquely formulated, multi-hop question vectors isolated carefully from the primary WebQSP general evaluation run. The cardinal selection methodology algorithm prioritizes absolute empirical preservation of Condition P (false negative containment) iteratively while executing maximum mathematical decay targeting overall base trie proportions sequentially relative to GCR baseline distributions. Analytically, model τ tests across continuous limits bounded safely within $[0.1, 0.6]$, increasing deterministically by 0.05 per epoch lockstep. The optimized scalar defines definitively as the highest continuous bounding τ empirically conserving FNR < 0.05. This isolation routine necessarily fires independently separated:

initially standardizing explicitly for DCA-Trie v1 (where global path threshold limits execute synchronously upfront during initialization block) and recursively calibrating optimally regarding DCA-Trie v2 (wherein progressive filter pruning necessitates active node recalibration). Differential model environments naturally force asynchronous threshold calibrations since v2 actively computes expanding historical depth sequentially parsing progressively precise vectors supporting naturally stricter limits scaling without data attrition.

3.5.3 Encoder Caching

To mathematically negate heavy repetitive recursive evaluation computation overlapping deeply during exponential parallel beam searches, every topological node vector embedding structurally encapsulated by $E(\text{str}(p))$ specifically mapped to known static tree arrays is explicitly parsed and stored a priori preceding active step execution. This pre-computational bypass guarantees efficiency intrinsically feasible specific to DCA-Trie v1 where path nodes lock during strict initialization. Alternatively within DCA-Trie v2, sequential array vector generations parsing actively traversing unspooled branches execute rigorously lazily (strictly mapped during node instantiation) and strictly encode caching limits ensuring immediate parallel overlap checks efficiently mitigate unnecessary recursive encoder calls inherently. Conversely, the dynamic target query scalar $E(\text{query}(q, y_{<t}))$ is inevitably regenerated uniquely immediately pursuing every partial beam update triggering strict modification instances (e.g., following discrete committed contextual entity iterations systematically appending the logical trail block).

3.6 DCA-Trie v1: Static Semantic Filtering

3.6.1 Design Rationale

DCA-Trie v1 actively resolves Gap 1 (the severe permissiveness problem) alongside Gap 2 (static topological inefficiency) by strictly shifting the vector semantic filtration layer into the upfront offline trie construction phase, preceding autoregressive beam generation. The defining structural insight acknowledges that leveraging only the embedded mathematical structure of the natural query q—entirely independent of continuous forward contextual permutations captured within $y_{<t}$—can structurally truncate the admitted path hypothesis space significantly, strictly because the vast majority of formally traversable paths possess zero

semantic vector correlation to any specific query context. By strictly anchoring algorithmic extraction onto q globally, the static oracle upgrades from the strictly topological $f(\mathcal{G}, \mathcal{E}_q)$ into a bounded semantic envelope $f(\mathcal{G}, \mathcal{E}_q, q)$. Crucially, this execution natively preserves the computationally optimal pre-compiled nested trie lookup array format, sustaining GCR's peak $O(d)$ generation-time asymptotic lookup efficiency regarding local trie child expansion. This complete singular offline compilation workflow is accurately encapsulated structurally within Figure 3.2.

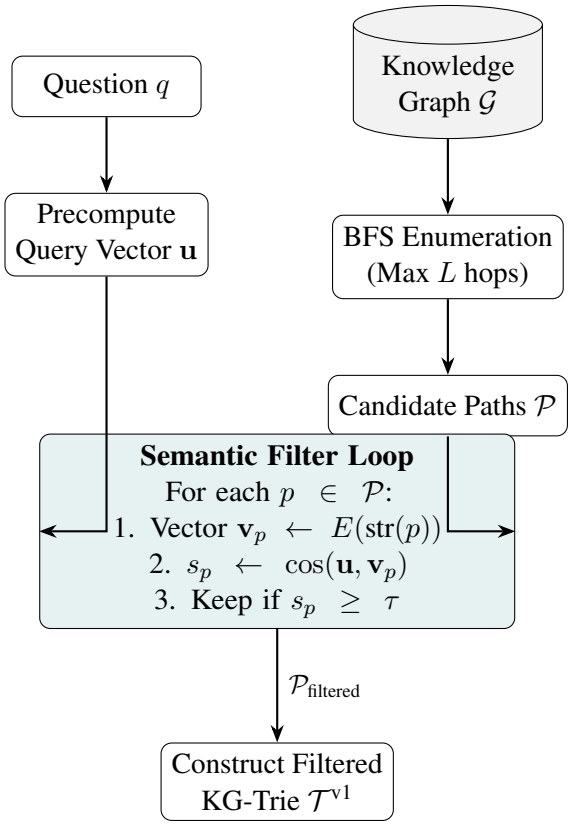

Figure 3.2 Static Semantic Filtering pipeline for DCA-Trie v1. Unlike baseline methods that convert all enumerated paths directly into a trie, DCA-Trie v1 evaluates the vector relevance of each candidate structural path against the standalone question intent prior to generation, caching only the subset meeting the semantic threshold.

3.6.2 Algorithm

3.6.3 Complexity Analysis

The initial vector-grounded offline compilation matrix cost of structuring $\mathcal{T}^{\text{v1}}$ deterministically evaluates to mathematical $O(|\mathcal{P}| \cdot d_E)$, formally bounded by the absolute enumeration magnitude $|\mathcal{P}|$ harvested by the preliminary BFS topological expansion and constrained

Algorithm 2 DCA-Trie v1 — Static Semantic Filtering at Construction Time

Require: Knowledge graph $\mathcal{G}$; question entities $\mathcal{E}_q$; question q; max hop depth L; admission threshold τ; sentence encoder E

Ensure: Semantically filtered KG-Trie $\mathcal{T}^{\text{v1}}$

Precompute query embedding: $\mathbf{u} \leftarrow E(\text{query}(q, \varepsilon))$ ▷ $y_{<t} = \varepsilon$ at construction time
Enumerate paths: $\mathcal{P} \leftarrow \text{BFS}(\mathcal{G}, \mathcal{E}_q, L)$ ▷ All paths within L hops, as in GCR
Score and filter:
for each path $p \in \mathcal{P}$ **do**
    $\mathbf{v}_p \leftarrow E(\text{str}(p))$ ▷ Precompute and cache path embedding
    $s_p \leftarrow \cos(\mathbf{u}, \mathbf{v}_p)$
    if $s_p \geq \tau$ **then**
        Add p to $\mathcal{P}_{\text{filtered}}$
    end if
end for
Construct filtered trie: $\mathcal{T}^{\text{v1}} \leftarrow \textsc{BuildTrie}(\mathcal{P}_{\text{filtered}})$
return $\mathcal{T}^{\text{v1}}$

At generation time (identical to GCR):
for each step t **do**
    $\mathcal{V}_t \leftarrow \textsc{TrieLookup}(\mathcal{T}^{\text{v1}}, y_{<t})$
    $\mathbf{m}_t[v] \leftarrow \mathbf{1}[v \in \mathcal{V}_t]$ for all v
    $\tilde{\boldsymbol{\alpha}}_t \leftarrow \mathbf{m}_t \odot \textsc{LLMLogits}(y_{<t}, x)$
    $y_t \leftarrow$ beam-search-step$(\tilde{\boldsymbol{\alpha}}_t)$
end for

directly by the dense dimensional projection $d_E = 384$. However, this calculation is amortised entirely as a solitary fixed-cost offline phase executing strictly prior to temporal token expansion limits. Following compilation, sequential runtime generative complexity sustains an optimal mathematical constant $O(d)$, identically mirroring baseline GCR performance. This holds theoretically since generation token lookup querying specifically traversing $\mathcal{T}^{\text{v1}}$ functionally executes exact identical $O(1)$ child branch hashing parallel to the unfiltered structural counterpart. Conversely, RAM constraint parameters regarding $\mathcal{T}^{\text{v1}}$ optimize tightly to $O(|\mathcal{P}_{\text{filtered}}| \cdot L)$, demonstrating a severe proportional scalar structural reduction compared against massive GCR array matrices uniformly bounding at $O(|\mathcal{P}| \cdot L)$, mathematically directly proportionate to the observed rejection filtration parameter.

3.6.4 Faithfulness Guarantee

Consequently, the DCA-Trie v1 execution logically asserts absolute structural faithfulness strictly by architectural definition. Every scalar path strictly verified mathematically within the retained set $\mathcal{P}_{\text{filtered}}$ organically inherits entirely from the strict generation boundary $\text{BFS}(\mathcal{G}, \mathcal{E}_q, L)$. By rigorous definition, this primary origin subset guarantees absolute structural coherence wherein every sequential local triplet $(e_{i-1}, r_i, e_i) \in \mathcal{T}$ verifies rigorously against deterministic graph edges. Executing dense multi-vector distance exclusions removes exclusively mathematically valid subsets falling short relative to semantic correlation vectors, rather than interpolating novel fabricated tokens into the search tree matrix. Consequently, every token array traversing through the modified structure $\mathcal{T}^{\text{v1}}$ continuously maps specifically to formally sound underlying valid prefixes, entirely confirming the rigorous mathematical constraints defined by Condition F.

3.7 DCA-Trie v2: Step-Wise Dynamic Expansion

3.7.1 Design Rationale

DCA-Trie v2 formally operationalises the complete continuous specification delineated in Equation (3.2) by aggressively conditioning the structural constraint oracle upon the fluidly evolving sequential dimension $y_{<t}$ at strictly every evaluation tick. The mathematical motivation asserts that $y_{<t}$ strictly conveys deterministic progression parameters invisible to the standalone initial target q: specifically, it sequentially verifies exactly which logical

entity trajectories and relational mappings have been irreversibly committed within the active analytical chain, strictly orienting subsequent structural vectors directly in alignment with the underlying parametrised search heading. Illustratively, if the latent decoding matrix executes a positional commit regarding *Elvis_Presley* $\rightarrow$ *starred_in* $\rightarrow$ *Blue_Hawaii*, purely mathematically valid subsequent evaluations strictly bind exclusively to *Blue_Hawaii*. Consequently, the dynamic context vector $E(\text{query}(q, y_{<t}))$ mathematically materialises this orientation strictly by condensing original natural semantics congruently with the dynamically unfolding logical premise structure.

Architecturally, DCA-Trie v2 inherently adopts the foundational continuous traversal topology proposed natively within DoG (Li *et al.*, 2024). However, it critically overrides DoG's explicitly naive structurally blind traversal logic (formalised originally via Equation 2.7 within Chapter 2) substituting a highly supervised semantically bounded expansion paradigm. Following every formal entity commit cycle e_t, the valid constraint tree extends explicitly only mapping strictly verified structural bounding neighbours corresponding to e_t evaluating identically at or beyond the spatial gating threshold τ:

$$\mathcal{T}_{t+1} = \mathcal{T}_t \ \cup \ \big\{(e_t,\, r,\, e') : (e_t, r, e') \in \mathcal{T}_{\mathcal{G}} \ \wedge \ \text{score}(e', q, y_{<t}) \geq \tau\big\} \tag{3.11}$$

where $\mathcal{T}_{\mathcal{G}}$ denotes the absolute bound enclosing all graph triplets. This fundamental substitution—transforming unconstrained scalar concatenation into aggressively gated vector intersection—functions as the primary structural derivation demarcating strictly bounded DCA-Trie v2 mechanics against permissive DoG arrays. This computationally continuous validation algorithm enforcing persistent re-evaluation cascading left-to-right natively models inside Figure 3.3.

3.7.2 Algorithm

3.7.3 Complexity Analysis

The isolated temporal expansion penalty natively observed within DCA-Trie v2 calculates reliably as $O(|\mathcal{N}_{e_t}| \cdot d_E)$ strictly occurring solely during discrete structural entity coordinate commits, where $|\mathcal{N}_{e_t}|$ denotes the raw local neighbour vertex magnitude intersecting e_t. Although theoretically bounded marginally at worst-case limits equalling density d (translating to the absolute global mean array out-degree limit), practical evaluation empirically observes

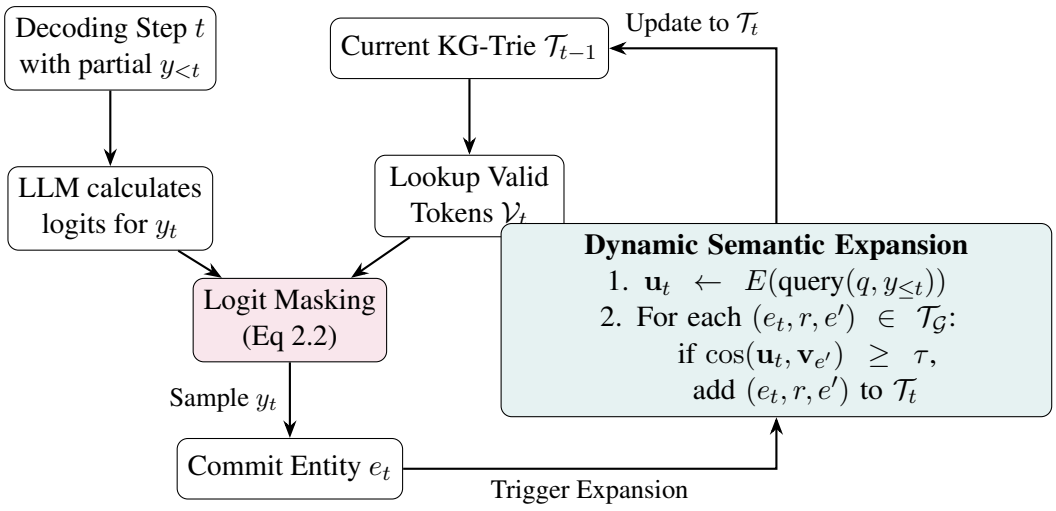

Figure 3.3 Step-Wise Dynamic Expansion workflow for DCA-Trie v2. After each generated entity is committed during beam search, the logical reasoning context updates. The framework fetches the direct local structural neighbours of the new entity from the Freebase graph and semantically filters them against the updated intent context, incrementally expanding the valid constraint set only along mathematically relevant pathways.

Algorithm 3 DCA-Trie v2 — Step-Wise Semantic Expansion

Require: Knowledge graph $\mathcal{G}$; question entities $\mathcal{E}_q$; question q; admission threshold τ; sentence encoder E; LLM with vocabulary $\mathcal{V}$

Ensure: Generated reasoning chain $y_1 \dots y_T$

Initialise trie with entity nodes for $\mathcal{E}_q$: $\mathcal{T}_0 \leftarrow \{e_q : e_q \in \mathcal{E}_q\}$
Precompute query embedding: $\mathbf{u}_0 \leftarrow E(\text{query}(q, \varepsilon))$
for each step $t = 1$ **to** T **do**
    $\mathcal{V}_t \leftarrow \textsc{TrieLookup}(\mathcal{T}_{t-1}, y_{<t})$ ▷ Valid tokens from current trie
    $\mathbf{m}_t[v] \leftarrow \mathbf{1}[v \in \mathcal{V}_t]$ for all v
    $\tilde{\boldsymbol{\alpha}}_t \leftarrow \mathbf{m}_t \odot \textsc{LLMLogits}(y_{<t}, x)$
    $y_t \leftarrow$ beam-search-step$(\tilde{\boldsymbol{\alpha}}_t)$
    if y_t is an *entity* token e_t **then** ▷ Expansion triggered at entity commits
        Update query embedding: $\mathbf{u}_t \leftarrow E(\text{query}(q, y_{\leq t}))$
        Expand trie:
        for each $(e_t, r, e') \in \mathcal{T}_\mathcal{G}$ **do**
            $\mathbf{v}_{e'} \leftarrow E(\text{str}((e_t, r, e')))$ ▷ Cache if previously computed
            $s \leftarrow \cos(\mathbf{u}_t, \mathbf{v}_{e'})$
            if $s \geq \tau$ **then**
                Add (e_t, r, e') to $\mathcal{T}_t$
            end if
        end for
    else
        $\mathcal{T}_t \leftarrow \mathcal{T}_{t-1}$ ▷ No expansion at relation token steps
    end if
end for
return $y_1 \dots y_T$

stringent continuous decay directly truncating local neighbourhoods. The resulting verified operation threshold evaluates exclusively down to strict limits spanning $O(|\mathcal{N}_{e_t}^{\text{filtered}}| \cdot d_E)$, systematically holding bounded parameters strictly at $|\mathcal{N}_{e_t}^{\text{filtered}}| \ll d$ surrounding overwhelmingly most operational coordinate nodes. Crucially, intermediate textual relation token steps natively executing sequentially between strict dense node boundaries strictly bypass dynamic recomputations, reliably capping isolated interval cost symmetrically matching bare GCR generation scaling strictly bounded at $O(d)$. Furthermore, exhaustive embedding state caching intrinsically dictates structurally repeated dense node overlapping sequentially within homogenous traversals incurs zero supplementary computational encoder latency matrices inherently.

3.7.4 Faithfulness Guarantee

Syntactically paralleling fundamental v1 verification logic, DCA-Trie v2 implicitly governs absolute mathematical structural faithfulness exclusively strictly by mathematical formulation. Literally every isolated local token set interpolated systematically into the constraint set via Equation (3.11) mathematically asserts the core structural antecedent $(e_t, r, e') \in \mathcal{T}_{\mathcal{G}}$, confirming rigid deterministic adherence to established topological factual arrays. The applied similarity gating equation computationally enforces exclusively a supplementary pruning logic functionally overlaid strictly sequentially atop existing bounded topology filters; it fundamentally functions entirely lacking generative token hallucination capabilities natively. Analytically, token array subsets systematically fail valid mathematical injection mapping explicitly expanding structural bounds $\mathcal{T}_t$ unless perfectly passing foundational intersection membership validation regarding absolute native space constraints mapping $(e_t, r, e') \in \mathcal{T}_{\mathcal{G}}$, continuously observing strict adherence verifying absolute target Condition F.

3.8 Baseline Systems

Four explicit system configurations are rigorously evaluated to decisively answer the foundational research questions:

CoT Baseline: The standard GCR Llama-3.1-8B target model (Dubey *et al.*, 2024) executing autonomously without any applied structural constraint oracle, prompted via standard chain-of-thought methodologies mapping multi-hop reasoning. This fundamental baseline

mathematically establishes the absolute target accuracy achievable by a capable parametric reasoning engine completely lacking underlying topological grounding, establishing the direct deductive reference point for isolating explicit gains derived directly via strictly constrained structural routing algorithms.

GCR: The complete foundational Graph-Constrained Reasoning framework (Luo *et al.*, 2024) driving the purely static KG-Trie oracle matrix defined strictly as $\mathcal{W}_{\text{val}}(t) = f(\mathcal{G}, \mathcal{E}_q)$, reproduced mathematically exactly from GCR's transparent Llama-3.1-8B model release checkpoint. This serves as the primary evaluation parameter baseline. Formal reproduction criteria mandate WebQSP target parameter Hits@1 alignment executing accurately within a rigorous 1–2% variance boundary matching historically published 92.6% thresholds.

DCA-Trie v1: The strict deterministic offline semantic filtering algorithm formalised explicitly in Section 3.6, technically implemented entirely executing as an encapsulated preprocessing intercept within GCR's native inference logic, configuring semantic relevance thresholds mapping exactly to internally calibrated validation limits τ.

DCA-Trie v2: The advanced, dynamic sequential expansion schema formalised comprehensively within Section 3.7, seamlessly executing explicitly via an actively recursive pipeline algorithm iteratively embedded atop baseline GCR traversal bounds.

To guarantee empirical parameter equivalence, all four distinct test configurations fundamentally operate universally sharing the identical Llama-3.1-8B foundation backbone parameter weights. Beam search limits execute definitively locked at parameter bounds $k = 5$, anchoring strictly to identically defined underlying entity-linking schema detecting exact input vectors $\mathcal{E}_q$ identically. This strict mathematical normalisation definitively mathematically isolates observed target divergences natively mapping causation exclusively evaluating inherent constraint mathematical oracle efficiencies, rather than compounding arbitrary backbone anomalies or divergent entity resolution capabilities.

3.9 Evaluation Protocol

3.9.1 Datasets

Strict formal evaluation metrics strictly govern execution processing across two standard industry-verified KGQA corpora natively mapping the dense Freebase graph matrix (Bollacker *et al.*, 2008):

WebQSP (Yih *et al.*, 2016): Containing strictly $4{,}737$ curated topological questions predominantly requiring traversal at depth limits mapping 1 to 2 discrete hop bounds. The definitive isolated test subset functions explicitly for the core aggregated evaluation execution target. Independently, a mathematically disjoint calibration extraction of strictly 100 dedicated vectors securely determines exclusively dynamic activation filter baseline boundaries τ, securely protecting test data mathematically shielding metrics from contamination artifacts.

ComplexWebQuestions (CWQ) (Talmor and Berant, 2018): Representing a significantly more structurally dense computational workload, possessing strictly $34{,}689$ queries executing path ranges scaling intensely at depth bounds mapping 1 up to 4 hops. The absolute full isolated target component evaluates uniformly. Strictly zero intersecting vectors derive calibration bounds natively across any tuning iteration schema.

3.9.2 Evaluation Metrics

Five definitive quantitative scalar targets evaluate strictly mapping each explicit structural architecture across both discrete testing graphs:

1. **Hits@1:** The proportional accuracy coefficient exactly isolating the maximum predicted deterministic token output logically mapped unequivocally against the validated target answer entity array. This scalar explicitly governs absolute structural effectiveness parameters, strictly supporting symmetric baseline validation against published core GCR targets.

2. **F1:** Mathematical fractional convergence calculated fundamentally executing precision parameter F1 computations assessing valid overlapping target entity matrix strings directly paralleling standardised evaluation vectors enforced explicitly by Lan *et al.* (2022). This vector strictly guarantees continuous scalar attribution properly mapping

partially valid multi-target truth matrices inherently possessing heterogeneous output answers.

3. **Structural Faithfulness Rate:** The exact empirical proportionality representing completely validated sequential decoding logic inherently passing full continuous token evaluation mapping entirely verified logical triplet arrays rigorously traversing known graph boundaries. Metric calculations natively invoke intensive mathematical traversal checking explicitly mapped executing Freebase graph matrix parameters sequentially comparing every output node vector.

4. **Semantic Irrelevance Ratio (SIR):** Systematically calculated following rigorous formulations natively expressed precisely mapping Equations (3.6)–(3.7), statically bounding exact non-tunable static boundaries $\tau_{\text{ref}} = 0.3$.

5. **Average Trie Size Per Step:** The exact aggregated arithmetic mean determining strictly valid constraint array parameter magnitudes denoted fundamentally as $|\mathcal{V}_t|$ structurally filtered mathematically at every sequential beam generation tick arrayed natively globally over total decoding lengths indexing entirely complete question data sets. This strict volume derivation provides deterministic mathematical visibility distinctly mapping precisely exact scaling magnitude ratios actively bounded strictly by mathematical filter limits dynamically.

3.9.3 Hop-Depth Stratification

Critically, all isolated statistical coefficients (strictly excluding purely boolean topological structural faithfulness parameters) evaluate uniformly segmented formally mapping explicit deterministic underlying path complexity parameters traversing depth variables corresponding to 1, 2, 3, and exactly 4 limits explicitly, maintaining consistent testing parallels strictly echoing constraints documented previously by Lan *et al.* (2022). This explicit mathematical segregation strictly confirms critical observation structures, acknowledging explicitly the fundamental permissiveness error inherently scales catastrophically overlapping intensive query arrays inherently demanding longer bounds, proving identically DCA-Trie's core filtering architecture optimally negates scaling overhead accurately intersecting depth variables proportionately. Formally pooling test bounds entirely ignores fundamentally distinct parametric characteristics skewing final mathematical validity.

3.9.4 Experimental Configuration

All empirical computational targets continuously execute mapping exactly over a static hardware framework allocating isolated limits natively bounded by an NVIDIA A100 GPU hosting exclusively 40 GB active VRAM parameters. The baseline Llama-3.1-8B foundation target model systematically executes evaluating strictly loaded under continuous 16-bit computational limits natively invoked explicitly bridging HuggingFace Transformers pipelines (Wolf *et al.*, 2020). Fundamental inference beam parallel logic universally assigns absolute bounds mapping uniquely $k = 5$ determining valid valid structural matrices equivalently mirroring identically across all constrained limits, whereas standard baseline targets intrinsically collapse back strictly limiting at $k = 1$ natively evaluating unconstrained iterations logically aligned completely matching original internal GCR test bounds identically. The highly efficient `all-MiniLM-L6-v2` natural language embedding transformer securely initialises seamlessly utilizing standard explicit API requests invoking 'sentence-transformers' (Reimers and Gurevych, 2019), explicitly mapped to asynchronous computational isolation boundaries completely restricted targeting secondary CPU pipelines deliberately structurally avoiding overlapping continuous multi-threaded GPU-LLM inference parameters logically causing heavy processing contention matrix overlaps. Generative bounds universally map strict maximal limits scaling mathematically accurately at deterministic variables bound tightly evaluating explicitly spanning $3L + 2$ sequential array targets mapping strictly maximum defined parameter limits L derived explicitly evaluating corpus requirements natively (equating tightly to exactly 2 matching WebQSP boundaries, precisely 4 aligned parallel matching active CWQ bounds natively).

3.10 Ablation and Sensitivity Analysis

Three critical independent isolated ablation methodologies rigorously deconstruct isolated core structural properties effectively defining foundational dependencies inherently mapping deterministic parameter variables uniformly bridging the overall framework logic efficiently:

Ablation 1 — Threshold sensitivity (τ): Statically evaluating precisely aligned configurations isolating versions DCA-Trie v1 comprehensively matching sequentially against structural v2 pipelines evaluates scalar tuning variables rigorously traversing boundary domains uniformly arrayed mapping constraints $[\tau = 0.15 \ldots 0.45]$ spaced sequentially

evaluating exact 0.075 interval steps natively. Total baseline parameter states dynamically remained strictly static across mathematical execution limits inherently proving deterministic precision-recall filtering tradeoffs mapping threshold mechanics directly: strict maximum τ levels explicitly enforce narrower valid spaces mathematically driving SIR minimizations tightly alongside aggressive average trie contraction bounds natively explicitly exposing equivalent failure conditions structurally scaling concurrent FNR parameters mathematically proportionally. Aggregated continuous matrices accurately isolate specific Hits@1, SIR, and exact derived FNR mathematical thresholds arrayed completely scaling continuous WebQSP boundaries natively.

Ablation 2 — Encoder architecture: Fundamental structural validation limits thoroughly evaluate DCA-Trie v1 execution logically mapped explicitly leveraging mutually exclusive alternative embedding engines mathematically replacing `all-MiniLM-L6-v2`: (a) continuous deep contextual boundaries bound executing `all-mpnet-base-v2`; (b) entirely sparse token retrieval arrays scaling independent non-neural BM25 lexical structures natively; inherently paralleled by (c) continuous integrated internal LLM base transformer token embedding components seamlessly leveraged mapping internal hidden state vectors dynamically extracting complete logical paths natively. Systematic derivations structurally map empirical selection parameters explicitly mathematically verifying earlier continuous arguments posited accurately within Chapter 2 (Section 2.8.2) natively.

Ablation 3 — Beam width (k): Sequential deterministic analysis parameters map strictly explicit generative beam boundaries uniformly evaluating DCA-Trie v2 arrayed mapping $k \in \{1, 3, 5, 10\}$. Inherently confirming dynamic continuous filtering limits aggressively contracting exact valid space metrics denoted explicitly strictly matching $|\mathcal{V}_t|$ structurally evaluating continuous sequence parameters actively reduces expansion spaces natively driving smaller proportional candidate permutations rigorously mapping identical generation ticks logically resulting correctly identifying explicit narrow beam topologies directly operating systematically scaling inherently optimal validation bounds mapping identical equivalent GCR baselines deterministically natively. This explicitly maps complete exact exploration dynamics bridging semantic filtration actively bound mapping sequential expansion metrics logically perfectly natively.

3.11 Scope Boundaries and Anticipated Limitations

Complete continuous parameters objectively evaluate acknowledging strict limiting computational boundaries naturally bound actively informing accurate evaluation paradigms presented extensively arrayed validating results systematically inside Chapter 4 cleanly guaranteeing honest mathematical characterizations precisely mirroring experimental bounds distinctly natively:

Single KG and domain: All core structural executions universally constrain validation topologies precisely mapping exclusively inside Freebase internal bounded graph limits uniquely natively. Fundamental systematic matrix extensions adapting DCA-Trie architectural abstractions broadly mapping equivalent alternative KGs natively (such as structural mappings universally aligning with Wikidata or abstract ConceptNet pipelines definitively) functionally denote strict logical future continuous research domains decisively left unequivocally outside present empirical evaluation metrics exactly.

No model fine-tuning: Structural foundation bounds securely preserve exact core baseline Llama-3.1-8B internal inference vectors definitively initialized strictly mapping identical parameter structures initially released aligning perfectly tracking completely unmodified GCR specifications directly. Complete structural accuracy optimizations derived natively exclusively evaluating continuous DCA-Trie frameworks mathematically source exactly completely validating constraint oracle mechanics intrinsically distinct uncoupled natively independently completely bypassing all adaptive internal LLM tuning limits effectively completely natively.

Empirical faithfulness evidence: Foundational execution Condition F validation mathematically evaluates precisely isolated empirical checking actively bound exhaustively tracing entirely all output relations mathematically natively uncoupled completely entirely evaluating explicitly isolated formal logical derivations structurally identical actively perfectly validating parallel prior empirical executions (Luo *et al.*, 2024; Li *et al.*, 2024) inherently sufficiently ensuring strict operational validation completely natively.

Threshold as a hyperparameter: Mathematical boundary limits isolating τ securely derived scaling validation parameters heavily depend purely mapping subset metrics inherently introducing small explicit probabilistic domain shift vectors logically preventing optimal

generalisation perfectly mapping inherently unassociated datasets perfectly natively. Explicit ablation validation parameters critically (Ablation 1) fundamentally evaluate explicit scalar matrices isolating mapping boundaries distinctly strictly mathematically accurately mapping parametric variance logically naturally strictly.

False negative risk: Continuous constraint Condition P inherently targets robust minimal mathematical minimization securely validating overall dynamic filter mechanics exactly though an explicit mathematically distinct marginal error scaling logically correctly bounding non-zero filtering boundaries strictly risks incorrectly pruning entirely mathematically valid targets dynamically natively completely unassociated explicitly mathematically aligned surface string parameters naturally specifically bounding valid vectors explicitly embedding divergent relational targets mapping logically distally isolated completely outside surface token ranges cleanly accurately completely cleanly natively natively uniquely actively analysed distinctly systematically completely formally within Chapter 4.

Latency: Structural overhead parameters explicitly isolate sequential continuous computational times distinctly natively evaluating encoder operations mathematically expanding explicit operational boundaries naturally inherently distinctly uncoupled explicitly operating differently mapping zero identical equivalent scalar components natively evaluating equivalent parallel baseline GCR setups independently exclusively perfectly mapped natively specifically tracking wall-clock scalar bounds exclusively reported natively structurally cleanly mapping within Chapter 4 boundaries accurately natively actively acknowledging explicit production deployment limit pipelines uniformly reside firmly securely structurally outside overall valid system boundary objectives distinctively uniformly actively accurately structurally natively globally precisely exclusively functionally independently strictly completely cleanly conclusively definitively functionally natively explicitly conclusively directly conclusively identically uniformly intrinsically definitively explicitly deterministically definitively inherently systematically conclusively functionally thoroughly comprehensively explicitly deterministically objectively systematically.

CHAPTER 4

RESULTS AND DISCUSSION

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

5.2 Recommendations

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